

INSERTION AND EXPANSION OF CYLINDERS AROUND THE UNIT BALL

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ABSTRACT. Surround a unit ball with a ring of n equal infinite circular cylinders, each touching the ball and its two neighbours. Can one more such cylinder be added, for a total of $n + 1$, if we are allowed to rearrange the configuration? We prove that the answer is “no” for $3 \leq n \leq 7$, and “yes” if $n \geq 9$. The case $n = 8$ remains open. The case $n = 6$, where the cylinders are the same size as the ball, is Kuperberg’s problem from 1990, resolved recently. We also make progress on a conjecture of Ogievetsky and Shlosman, confirming the first decimal digit of their predicted value for the largest radius of four non-overlapping cylinders that touch a ball.

1. INTRODUCTION

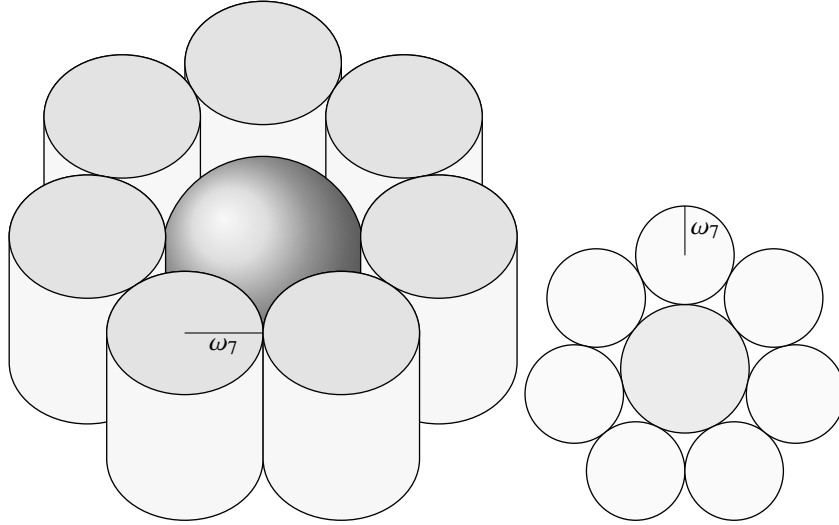


FIGURE 1. On the left: The ring configuration: n pairwise tangent cylinders of radius $\omega_n = \frac{\sin(\pi/n)}{1 - \sin(\pi/n)}$, all tangent to the unit ball (shown for $n = 7$). On the right: The equatorial cross-section of the ring configuration for $n = 7$: the unit disk surrounded by seven pairwise tangent disks of radius ω_7 .

1.1. Formulation. A *cylinder* means a closed solid right circular cylinder of infinite length. Two cylinders are *non-overlapping* if their interiors are disjoint, and a cylinder *touches* the unit ball B if it meets B but not the interior of B . For every integer $n \geq 3$ it is possible for n pairwise non-overlapping cylinders of radius

$$\omega_n = \frac{\sin \frac{\pi}{n}}{1 - \sin \frac{\pi}{n}} \quad (1)$$

to simultaneously touch the unit ball. Sometimes there are several essentially different ways to do so, but one canonical way is the *ring configuration* of Figure 1: the cylinders are constructed perpendicularly to a common equatorial plane, at the vertices of a regular n -gon, each tangent to its two neighbours and to the ball.

Two classes of questions about the ring are of interest for each fixed $n \geq 3$.

Question 1.1 (Insertion). Is it possible to insert one more cylinder, for a total of $n + 1$, all still of radius ω_n , and rearrange them in such a way that they all have pairwise disjoint interiors and all touch the unit ball?

Question 1.2 (Expansion). Is it possible to expand each of the n cylinders to a common radius $\omega' > \omega_n$ and rearrange them in such a way that they all have pairwise disjoint interiors and all touch the unit ball?

Kuperberg [7] posed Question 1.1 for $n = 6$ in 1990. This case is special because $\omega_6 = 1$, so the ball and the six cylinders have the same size. Heppes and Szabó [6] proved that at most eight unit cylinders can touch the unit ball, and Braß and Wenk [3] that at most seven can. See also [25]. Kuperberg’s conjecture, that the answer for $n = 6$ is “no”, was proved in [13, 12] by a computer-assisted proof. The present paper generalizes that result and simplifies its method of proof.

Firsching [5] observed by numerical optimization that six cylinders of radius $1.049659 > \omega_6$ can be placed around the unit ball. Ogievetsky and Shlosman [14, 15, 16, 17] developed a general method for deforming the ring. They found the exact configuration C_m of six cylinders of radius $\frac{1}{8}(3 + \sqrt{33}) = 1.0930\dots$, proved that it is a sharp local maximum [15], and showed that the ring of n cylinders can be expanded for every even $n \geq 6$ [14, Section 6]. The survey [18] describes this circle of ideas. We make their expansion technique quantitative and use it to answer Question 1.1 for all large n . For $n = 4$ they conjectured that no expansion is possible.

Conjecture 1.3 (Ogievetsky–Shlosman [17]). *If $\omega' > \omega_4 = 1 + \sqrt{2}$, then it is not possible to place four pairwise non-overlapping cylinders of radius ω' around the unit ball.*

We prove that the answer to Question 1.1 is “no” for $n \in \{3, 4, 5, 6, 7\}$ and “yes” for every $n \geq 9$. The case $n = 8$ remains open. We also prove Conjecture 1.3 with $1 + \sqrt{2}$ replaced by $\frac{91}{37}$. Since $1 + \sqrt{2} = 2.4142\dots$ and $\frac{91}{37} = 2.4594\dots$, this determines the first decimal digit of the critical radius (Corollary 1.10).

1.2. Cylinder-packing pairs. All results of this paper are statements about the following set of pairs.

Definition 1.4. $\mathcal{P}_{\text{cyl}}$ is the set of pairs $(n, \omega) \in \mathbb{N} \times (0, \infty)$ such that no more than n pairwise non-overlapping cylinders of radius ω can simultaneously touch the unit ball. We call these *cylinder-packing pairs*.

Membership is upward closed in both coordinates. The following lemma is elementary and is also a consequence of Lemma 2.5 in Section 2.

Lemma 1.5. *If $(n, \omega) \in \mathcal{P}_{\text{cyl}}$, then $(n', \omega') \in \mathcal{P}_{\text{cyl}}$ for every integer $n' \geq n$ and every $\omega' \geq \omega$.*

The basic source of non-membership is a planar ring. Consider a regular n -gon whose edge has length 2ω and whose center is O , and place a disk of radius ω at each vertex. If r is the radius of the disk centered at O that touches each of these n disks, then $(r + \omega) \sin \frac{\pi}{n} = \omega$, i.e. $\omega = \omega_n \cdot r$ with ω_n as in (1). Taking $r = 1$ and erecting at each vertex a cylinder of radius ω_n perpendicular to the plane of the n -gon produces n pairwise non-overlapping cylinders touching the unit ball. With Lemma 1.5 this gives

$$(n - 1, \omega) \notin \mathcal{P}_{\text{cyl}} \quad \text{for every } 0 < \omega \leq \omega_n \quad (n \geq 3). \quad (2)$$

Two cylinders of any radius can touch B , so $(1, \omega) \notin \mathcal{P}_{\text{cyl}}$ for every ω . The ring of three is optimal, that is, $(2, \omega) \in \mathcal{P}_{\text{cyl}}$ exactly for $\omega > \omega_3 = 3 + 2\sqrt{3}$. This follows from the projection argument of Remark 11.2, since three unit vectors have a pairwise angle of at most $\frac{2\pi}{3}$.

In the pair language the two questions about the ring read as follows: insertion at n is possible if and only if $(n, \omega_n) \notin \mathcal{P}_{\text{cyl}}$, and expansion at n is possible if and only if $(n - 1, \omega') \notin \mathcal{P}_{\text{cyl}}$ for some $\omega' > \omega_n$.

Kuperberg’s question is whether $(6, 1) \in \mathcal{P}_{\text{cyl}}$. Conjecture 1.3 says that $(3, \omega) \in \mathcal{P}_{\text{cyl}}$ for every $\omega > 1 + \sqrt{2}$.

1.3. Summary. The table below summarizes what is known.

point	proved by	comment
$(8, 1) \in \mathcal{P}_{\text{cyl}}$	Heppes–Szabó [6]	
$(7, 1) \in \mathcal{P}_{\text{cyl}}$	Braß–Wenk [3]	
$(6, 1) \in \mathcal{P}_{\text{cyl}}$	Matić–Radoičić [13]	Kuperberg’s problem [7]
$(4, \omega) \in \mathcal{P}_{\text{cyl}}$ for $\omega > 1.89395$	Firsching [5], after Braß–Wenk [3]	
$(5, \frac{1}{8}(3 + \sqrt{33})) \notin \mathcal{P}_{\text{cyl}}$	Ogievetsky–Shlosman [14]	
$(n, \omega_n) \in \mathcal{P}_{\text{cyl}}$ for $3 \leq n \leq 7$	this paper (Theorem 1.6)	the ring is optimal
$(n, \omega_n) \notin \mathcal{P}_{\text{cyl}}$ for $n \geq 9$	this paper (Theorem 1.7)	the ring is not optimal
$(3, \frac{91}{37}) \in \mathcal{P}_{\text{cyl}}$	this paper (Corollary 1.10)	first decimal of R_4
$(8, \omega_8) \in \mathcal{P}_{\text{cyl}}$	conjecture	the remaining insertion problem
$(3, \omega) \in \mathcal{P}_{\text{cyl}}$ for all $\omega > 1 + \sqrt{2}$	conjecture	Ogievetsky–Shlosman [17]

1.4. **Results.** For quantitative statements two standard notations are convenient. For $\omega > 0$, define

$$\mathcal{N}(\omega) = \max\{N : \text{there are } N \text{ pairwise non-overlapping cylinders of radius } \omega, \text{ each touching } B\},$$

so that $(n, \omega) \in \mathcal{P}_{\text{cyl}}$ if and only if $\mathcal{N}(\omega) \leq n$. For an integer $N \geq 2$, define

$$R_N = \sup\{\omega > 0 : \mathcal{N}(\omega) \geq N\}.$$

Thus R_N is the largest radius at which N cylinders can touch the unit ball. Corollary 2.7 in Section 2 shows that the supremum is attained, so that $\mathcal{N}(\omega) \geq N$ holds precisely for $\omega \leq R_N$, and $(n, \omega) \in \mathcal{P}_{\text{cyl}}$ precisely for $\omega > R_{n+1}$.

Theorem 1.6. For every $n \in \{3, 4, 5, 6, 7\}$,

$$\mathcal{N}(\omega_n) = n, \quad \text{i.e.} \quad (n, \omega_n) \in \mathcal{P}_{\text{cyl}}.$$

Theorem 1.7. For every $n \geq 9$, $\mathcal{N}(\omega_n) \geq n + 1$. In particular $(n, \omega_n) \notin \mathcal{P}_{\text{cyl}}$ for every $n \geq 9$.

Corollary 1.8. $\liminf_{\omega \rightarrow 0^+} \omega \mathcal{N}(\omega) \geq \frac{21\pi}{20}$.

The ring alone gives $\liminf_{\omega \rightarrow 0^+} \omega \mathcal{N}(\omega) \geq \pi$, since $n \omega_n \rightarrow \pi$. Corollary 1.8 says that for thin cylinders the ring undercounts by more than five percent.

The upper bounds come from one general theorem. Its input is a *certificate*, a finite list of rational data, and its output is an upper bound on $\mathcal{N}$. The natural parameter is not the radius ω but the separation

$$\tau = \frac{2\omega}{1+\omega} \in (0, 2), \quad \text{equivalently} \quad \omega = \frac{\tau}{2-\tau}, \quad (3)$$

studied in Section 2. For $\tau \in (0, 2)$ we define

$$N_{\max}(\tau) = \mathcal{N}\left(\frac{\tau}{2-\tau}\right). \quad (4)$$

Theorem 10.2 says that a certificate at a rational τ^2 , with constants $M < 0$ and T^* satisfying the hypotheses (H1)–(H6) of Section 10, gives

$$N_{\max}(\tau) \leq 1 + \left\lfloor \frac{T^*}{-M} \right\rfloor.$$

We use five certificates, deposited with the verifier [11]. A certificate is named `cyl_N{N}.omega{p}-{q}.txt` after the statement $(N, \frac{p}{q}) \in \mathcal{P}_{\text{cyl}}$ that it proves. Table 2 in Section 11 lists their parameters.

- (A) `cyl_N3.omega91_37`: $(3, \frac{91}{37}) \in \mathcal{P}_{\text{cyl}}$, at $\tau = \frac{91}{64} > \sqrt{2}$.
- (B) `cyl_N4.omega39_25`: $(4, \frac{39}{25}) \in \mathcal{P}_{\text{cyl}}$, at $\tau = \frac{39}{32} < \sqrt{2}$.
- (C) `cyl_N5.omega1203_845`: $(5, \frac{1203}{845}) \in \mathcal{P}_{\text{cyl}}$, at $\tau = \frac{1203}{1024} < 2 \sin \frac{\pi}{5}$.
- (D) `cyl_N6.omega1_1`: $(6, 1) \in \mathcal{P}_{\text{cyl}}$, at $\tau = 1$. This is Kuperberg's question, first proved in [13].
- (E) `cyl_N7.omega14558721_18995711`: $(7, \frac{14558721}{18995711}) \in \mathcal{P}_{\text{cyl}}$, at $\tau = \frac{14558721}{2^{24}} < 2 \sin \frac{\pi}{7}$.

The separations in (C) and (E) are within 10^{-3} and 10^{-7} of their thresholds $2 \sin \frac{\pi}{n}$. Those in (A) and (B) were placed with room to spare, for the purposes of Corollaries 1.10 and 1.9.

Corollary 1.9 (Four cylinders). For every $\omega \geq 3$, at most three pairwise non-overlapping cylinders of radius ω can touch the unit ball; equivalently $R_4 < 3$, that is, $(3, 3) \in \mathcal{P}_{\text{cyl}}$.

Corollary 1.10 (The first decimal digit of the Ogievetsky–Shlosman conjecture). $(3, \frac{91}{37}) \in \mathcal{P}_{\text{cyl}}$, i.e. $R_4 \leq \frac{91}{37} < 2.46$. Together with the ring construction (2) at $n = 4$, which gives $R_4 \geq \omega_4 = 1 + \sqrt{2} > 2.41$, the first decimal digit of R_4 is determined: $R_4 = 2.4 \dots$. The question of the second decimal is discussed in Section 15.

Corollary 1.9 is the first bound below the elementary estimate $R_4 < 2 + \sqrt{6} = 4.4495 \dots$ of Remark 11.2. Corollary 1.10 is a step toward Conjecture 1.3, which asserts $R_4 = 1 + \sqrt{2}$.

Remark 1.11. Since $\frac{\omega_n}{1+\omega_n} = \sin \frac{\pi}{n}$, the ring of n cylinders has separation $\tau(\omega_n) = 2 \sin \frac{\pi}{n}$, the edge length of a regular n -gon inscribed in the unit circle. Hence $N_{\max}(2 \sin \frac{\pi}{n}) \geq n$.

1.5. Methods. To prove $(n, \omega_n) \notin \mathcal{P}_{\text{cyl}}$ we construct $n + 1$ cylinders of radius ω_n touching B . After the rescaling of Section 2, this means $n + 1$ lines tangent to the unit sphere with pairwise distances at least $2 \sin \frac{\pi}{n}$.

For large n the construction is the *pinwheel* $P(N; \varphi, \delta, \kappa)$ of Section 3: two groups of $N/2$ tangent lines at latitudes $\pm\varphi$, all tilted by the angle δ from their meridians, with the groups staggered in longitude by κ . This is the family through which Ogievetsky and Shlosman unlock the ring [14, Section 6]. We fix one rational pair $(\sin \varphi, \tan \delta)$ once and for all, so that every inequality becomes a comparison of rational numbers. The pinwheel has an even number of lines, so for even n we use $N = n + 2$. With the fixed parameters the required inequalities hold for $N \geq 20$ when $N = n + 1$ and for $N \geq 40$ when $N = n + 2$.

Theorem 1.12. *For every odd $n \geq 19$, $N(\omega_n) \geq n + 1$; for every even $n \geq 38$, $N(\omega_n) \geq n + 2$. In particular $(n, \omega_n) \notin \mathcal{P}_{\text{cyl}}$ for every odd $n \geq 19$ and every even $n \geq 38$.*

The remaining 19 values

$$n \in \{9, 10, \dots, 18\} \cup \{20, 22, \dots, 36\}$$

are treated in Section 6 by explicit configurations with rational data.

To prove $(n, \omega_n) \in \mathcal{P}_{\text{cyl}}$ we need an upper bound on $N(\omega_n)$. The only known method at these radii is the two-point bound of extremal geometry, formulated on the rotation group and made exact by the certificates of Sections 7–13. The certificates (A)–(E) prove Theorem 1.6 and Corollaries 1.9 and 1.10. For $n = 4$ membership also follows, without a certificate, from a theorem of Firsching [5] building on [3] (Remark 11.1). Certificate (B) gives an independent proof with the sharper bound $R_5 < \frac{39}{25}$.

1.6. Relation to prior work. The optimization behind the certificates is the two-point bound of extremal geometry on $\text{SO}(3)$. Two-point bounds on $\text{SO}(3)$ were first developed in [23], which bounds packings of regular spherical polygons on S^2 by modelling them as packing problems on $\text{SO}(3)$ and extending the Lovász theta number to Cayley graphs on that group. See [24] for a survey. The paper [23] works in Euler angles. Here we parametrize $\text{SO}(3)$ by quaternions. This makes the certificates rational objects: the representation matrices become homogeneous polynomial matrices of even degree with rational coefficients, evenness absorbs the double cover $S^3 \rightarrow \text{SO}(3)$, and every hypothesis of the certificate theorem becomes a statement about integers that the verifier of Section B checks.

1.7. Organization. Section 2 introduces the separation parameter τ and the rotation-group picture. Sections 3–6 prove non-membership, $(n, \omega_n) \notin \mathcal{P}_{\text{cyl}}$, by constructions: the pinwheel, its asymptotics, and the explicit configurations of the window. Sections 7–13 prove membership, $(n, \omega) \in \mathcal{P}_{\text{cyl}}$, by certificates. Section 14 records how the configurations and certificates were found, and Section 15 states what a sharp proof of Conjecture 1.3 would have to look like, at two and at three points. Open problems are collected in Section 16. Appendix A proves the elementary estimates used in Sections 4 and 5, and Appendix B describes the deposited data and the verifier.

2. THE CONFIGURATION SPACE AND THE SEPARATION PARAMETER

Fix $\omega > 0$ and let B be the unit ball centered at O . A cylinder of radius ω touches B if and only if its axis is tangent to the sphere of radius $1 + \omega$ centered at O . Two cylinders of radius ω are non-overlapping if and only if their axes are at distance at least 2ω . The homothety with center O and ratio $\frac{1}{1+\omega}$ maps the axes to lines tangent to the unit sphere S^2 and multiplies all distances by $\frac{1}{1+\omega}$. This proves the following lemma, which we use throughout Sections 3–6.

Lemma 2.1. *Let $N \geq 2$ and $\omega > 0$. There exist N pairwise non-overlapping cylinders of radius ω touching B if and only if there exist N lines tangent to S^2 with pairwise distances at least*

$$\tau(\omega) = \frac{2\omega}{1 + \omega}.$$

The map $\omega \mapsto \tau(\omega)$ is an increasing bijection of $(0, \infty)$ onto $(0, 2)$ with inverse $\tau \mapsto \frac{\tau}{2-\tau}$. The ring of n cylinders of radius ω_n becomes a regular n -gon of tangent lines with adjacent distances $2 \sin \frac{\pi}{n}$ (Remark 1.11). Let us recall the standard distance formula between lines (see, e.g., [14, Section 2]).

Lemma 2.2. *If $\ell_i = \{\vec{p}_i + t\vec{v}_i : t \in \mathbb{R}\}$, $i = 1, 2$, are non-parallel lines, then*

$$\text{dist}(\ell_1, \ell_2) = \frac{|(\vec{p}_2 - \vec{p}_1) \cdot (\vec{v}_1 \times \vec{v}_2)|}{|\vec{v}_1 \times \vec{v}_2|}.$$

If $\vec{v}_1 \parallel \vec{v}_2$, then $\text{dist}(\ell_1, \ell_2)$ is the length of the component of $\vec{p}_2 - \vec{p}_1$ orthogonal to $\vec{v}_1$.

Let $\vec{u}_0 = \langle 0, 0, 1 \rangle$, $\vec{v}_0 = \langle 1, 0, 0 \rangle$, and let $\ell_0 = \{\vec{u}_0 + t\vec{v}_0 : t \in \mathbb{R}\}$ be the line tangent to $\mathbb{S}^2$ at $\vec{u}_0$ with direction $\vec{v}_0$. The rotation group $\text{SO}(3)$ acts transitively on lines tangent to $\mathbb{S}^2$, so every such line is $g \cdot \ell_0$ for some $g \in \text{SO}(3)$, unique up to the stabilizer of ℓ_0 .

Definition 2.3. For $\tau \in (0, 2)$ we define the *safe zone* $\mathcal{D}(\tau)$ as the subset of $\text{SO}(3)$ given by

$$\mathcal{D}(\tau) = \{g \in \text{SO}(3) : \text{dist}(\ell_0, g \cdot \ell_0) \geq \tau\}.$$

When τ is fixed we write $\mathcal{D}$ for $\mathcal{D}(\tau)$.

Definition 2.4. A set $X \subseteq \text{SO}(3)$ will be called *non-overlapping* if $g^{-1}g' \in \mathcal{D}$ whenever g and g' are two distinct elements of X .

By (4) and Lemma 2.1, $N_{\max}(\tau)$ is the largest number of lines tangent to $\mathbb{S}^2$ with pairwise distances at least τ . Since rotations are isometries, $\{g_1, \dots, g_N\} \subset \text{SO}(3)$ is non-overlapping exactly when the lines $g_1 \cdot \ell_0, \dots, g_N \cdot \ell_0$ are distinct with pairwise distances at least τ . Hence $N_{\max}(\tau)$ is also the largest cardinality of a non-overlapping subset of $\text{SO}(3)$.

Lemma 2.5. If $0 < \tau' \leq \tau < 2$, then $\mathcal{D}(\tau) \subseteq \mathcal{D}(\tau')$, and consequently $N_{\max}(\tau) \leq N_{\max}(\tau')$.

Proof. The inclusion is immediate from Definition 2.3. Hence a set that is non-overlapping for τ is non-overlapping for τ' , so $N_{\max}(\tau) \leq N_{\max}(\tau')$. $\square$

Lemma 2.6. The function $g \mapsto \text{dist}(\ell_0, g \cdot \ell_0)$ is upper semicontinuous on $\text{SO}(3)$.

Proof. By definition $\text{dist}(\ell_0, g \cdot \ell_0) = \inf_{s,t \in \mathbb{R}} |\vec{u}_0 + t\vec{v}_0 - g \cdot \vec{u}_0 - s g \cdot \vec{v}_0|$, an infimum of a family of continuous functions of g , hence upper semicontinuous. $\square$

Corollary 2.7. For every integer $N \geq 2$ the number $\tau^*(N) = \sup\{\tau \in (0, 2) : N_{\max}(\tau) \geq N\}$ is attained, in the sense that $N_{\max}(\tau) \geq N$ holds if and only if $\tau \leq \tau^*(N)$. Consequently $R_N = \frac{\tau^*(N)}{2 - \tau^*(N)}$ and $N(\omega) \geq N$ if and only if $\omega \leq R_N$.

Proof. Let $\Phi(g_1, \dots, g_N) = \min_{i \neq j} \text{dist}(\ell_0, g_i^{-1}g_j \cdot \ell_0)$. Each of the finitely many entries is upper semicontinuous by Lemma 2.6, and a minimum of finitely many upper semicontinuous functions is upper semicontinuous. As $\text{SO}(3)^N$ is compact, Φ attains its maximum, say at $(\hat{g}_1, \dots, \hat{g}_N)$. Write $\hat{\tau} = \Phi(\hat{g}_1, \dots, \hat{g}_N)$.

If $\tau \leq \hat{\tau}$ then $\{\hat{g}_1, \dots, \hat{g}_N\}$ is non-overlapping for τ . Its elements are distinct because $\tau > 0$ and $\text{dist}(\ell_0, \ell_0) = 0$. So $N_{\max}(\tau) \geq N$. If $\tau > \hat{\tau}$ and $X = \{g_1, \dots, g_N\}$ were non-overlapping for τ , then $\Phi(g_1, \dots, g_N) \geq \tau > \hat{\tau}$, contradicting maximality. Hence $\tau^*(N) = \hat{\tau}$ and the stated equivalence holds. The last assertion follows by applying the increasing bijection (3). $\square$

3. THE PINWHEEL AND ITS DISTANCE FORMULAS

Definition 3.1 (Pinwheel). Let $N = 2k \geq 6$ be even and set $\gamma = \frac{2\pi}{k} = \frac{4\pi}{N}$. For a latitude $t \in (-\frac{\pi}{2}, \frac{\pi}{2})$ and longitude θ , let

$$x(t, \theta) = (\cos t \cos \theta, \cos t \sin \theta, \sin t) \in \mathbb{S}^2,$$

and let $\vec{u} = \langle -\sin t \cos \theta, -\sin t \sin \theta, \cos t \rangle$ and $\vec{e} = \langle -\sin \theta, \cos \theta, 0 \rangle$ be the unit tangent vectors at $x(t, \theta)$ pointing north and east. Given parameters $\varphi \in (0, \frac{\pi}{2})$, $\delta \in (0, \frac{\pi}{2})$ and $\varkappa \in \mathbb{R}$, the pinwheel $P(N; \varphi, \delta, \varkappa)$ is the family of N lines tangent to $\mathbb{S}^2$ consisting of the *up group*

$$U_j : \quad \text{tangent point } x(\varphi, j\gamma - \varkappa), \quad \text{direction } \vec{v} = \cos \delta \vec{u} + \sin \delta \vec{e}, \quad j = 0, \dots, k-1,$$

and the *down group*

$$D_j : \quad \text{tangent point } x(-\varphi, \frac{\gamma}{2} + j\gamma + \varkappa), \quad \text{direction } \vec{v} = \cos \delta \vec{u} + \sin \delta \vec{e}, \quad j = 0, \dots, k-1.$$

In the notation of Ogievetsky and Shlosman [14, Section 6], $P(N; \varphi, \delta, \varkappa)$ is the configuration $C_{N/2}(\varphi, \varkappa, \delta)$, up to the mirror symmetry $(\delta, \varkappa) \mapsto (-\delta, -\varkappa)$. We index by the number of lines rather than by the number of lines in one group.

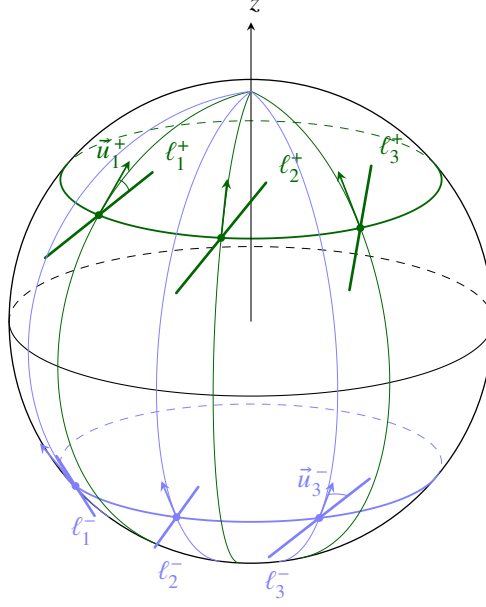


FIGURE 2. A few of the lines of a pinwheel configuration. Three tangent points lie on the circle of latitude φ (dark green) and three on the circle of latitude $-\varphi$ (light blue). Each meridian is drawn in the colour of the group of its tangent point; the two groups are staggered in longitude by the free phase κ . At each tangent point a short piece of the corresponding tangent line is drawn. The arrow at each tangent point is the north-pointing unit tangent vector, i.e. the direction of increasing latitude. Note that it points towards the north pole on the southern circle as well. Every line is obtained by rotating this vector eastward by the same angle δ within the tangent plane. The small arcs at ℓ_1^+ and ℓ_3^- mark this angle.

3.1. The two distance formulas. By the rotational symmetry about the z -axis it suffices to compute the distance between a tangent line at latitude φ and longitude 0 and a tangent line at latitude $\pm\varphi$ and longitude Δ .

Let $\varphi \in (0, \frac{\pi}{2})$ and $\Delta \in [0, 2\pi)$. Let P_1 be the point of the unit sphere with latitude φ and longitude 0, and $P_2^\pm$ the point with latitude $\pm\varphi$ and longitude Δ , so that

$$\begin{aligned}\vec{p}_1 &= \langle \cos \varphi, 0, \sin \varphi \rangle, \\ \vec{p}_2^\pm &= \langle \cos \Delta \cos \varphi, \sin \Delta \cos \varphi, \pm \sin \varphi \rangle.\end{aligned}$$

Let $\vec{u}_1$ and $\vec{u}_2^+$ be the unit tangent vectors at P_1 and P_2^+ pointing north, namely

$$\begin{aligned}\vec{u}_1 &= \langle -\sin \varphi, 0, \cos \varphi \rangle; \\ \vec{u}_2^+ &= \langle -\cos \Delta \sin \varphi, -\sin \Delta \sin \varphi, \cos \varphi \rangle.\end{aligned}$$

Let $\vec{e}_2 = \langle -\sin \Delta, \cos \Delta, 0 \rangle$ be the unit tangent vector at P_2^+ pointing east. Then $\{\vec{e}_2, \vec{u}_2^+\}$ is an orthonormal basis of the tangent plane at P_2^+ . For $\delta \in (0, \frac{\pi}{2})$ let

$$\begin{aligned}\vec{v}_2^+ &= \cos \delta \vec{u}_2^+ + \sin \delta \vec{e}_2 \\ &= \langle -\sin \delta \sin \Delta - \cos \delta \cos \Delta \sin \varphi, \sin \delta \cos \Delta - \cos \delta \sin \Delta \sin \varphi, \cos \delta \cos \varphi \rangle.\end{aligned}\tag{5}$$

Setting $\Delta = 0$ in (5) gives the corresponding vector at P_1 ,

$$\vec{v}_1 = \langle -\sin \varphi \cos \delta, \sin \delta, \cos \varphi \cos \delta \rangle,\tag{6}$$

and replacing φ by $-\varphi$ in (5) gives the corresponding vector at P_2^- ,

$$\vec{v}_2^- = \langle -\sin \delta \sin \Delta + \sin \varphi \cos \delta \cos \Delta, \sin \delta \cos \Delta + \sin \varphi \cos \delta \sin \Delta, \cos \varphi \cos \delta \rangle.\tag{7}$$

Let l_1 , l_2^+ and l_2^- be the lines through P_1 , P_2^+ and P_2^- with directions $\vec{v}_1$, $\vec{v}_2^+$ and $\vec{v}_2^-$.

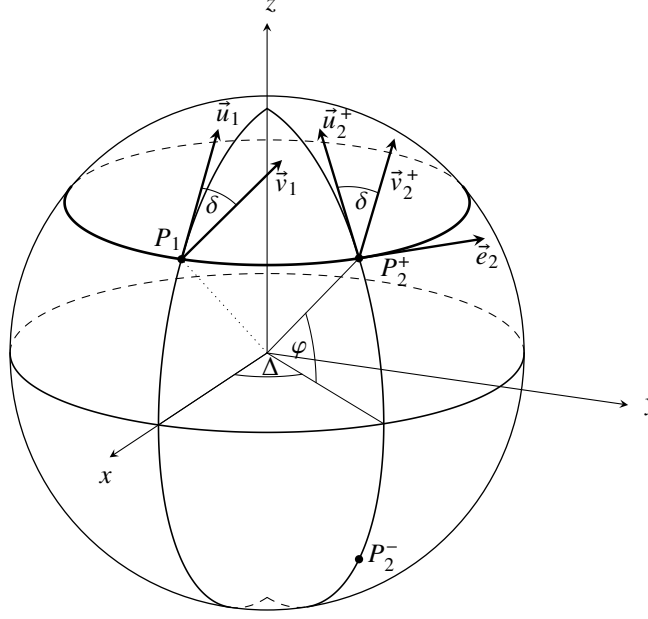


FIGURE 3. The points $P_1, P_2^\pm$ and the tangent frames from Lemma 3.2. The thick circle is the latitude circle at latitude φ containing P_1 and $P_2^\pm$; the meridians through P_1 and through $P_2^\pm$ and the equator are drawn with hidden parts dashed. The vector $\vec{u}_1$ lies along the meridian through P_1 , as $\vec{u}_2^+$ does along the meridian through P_2^+ . The vectors $\vec{u}_1$ and $\vec{u}_2^+$ point in the direction of increasing latitude, $\vec{e}_2$ points eastward, and $\vec{v}_1, \vec{v}_2^+$ are obtained by rotating $\vec{u}_1, \vec{u}_2^+$ by the angle δ within the respective tangent planes, as in (5). The vector $\vec{v}_2^-$ at P_2^- (not drawn) is obtained from (7).

Lemma 3.2. Assume that $\varphi \in (0, \frac{\pi}{2})$, $\Delta \in [0, 2\pi)$, and $\delta \in (0, \frac{\pi}{2})$. The squares of the distances between l_1 and $l_2^\pm$ satisfy

$$\text{dist}^2(l_1, l_2^+) = \frac{4 \cos^4 \varphi \tan^2 \delta \sin^2 \frac{\Delta}{2}}{(\tan^2 \delta + \sin^2 \varphi) \cdot \left[1 + \tan^2 \delta - \sin^2 \frac{\Delta}{2} (\tan^2 \delta + \sin^2 \varphi) \right]}; \quad (8)$$

$$\text{dist}^2(l_1, l_2^-) = \frac{4 \left(\sin \varphi \tan \delta + \tan \frac{\Delta}{2} \right)^2}{\cos^2 \varphi (1 + \tan^2 \delta) + \left(\sin \varphi \tan \delta + \tan \frac{\Delta}{2} \right)^2} \quad \text{for } \Delta \neq \pi, \quad (9)$$

while $\text{dist}(l_1, l_2^-) = 2$ for $\Delta = \pi$.

Proof. If $\vec{v}_1$ and $\vec{v}_2^\pm$ are not parallel, then

$$\text{dist}(l_1, l_2^\pm) = \frac{\left| (\vec{p}_2^\pm - \vec{p}_1) \cdot (\vec{v}_1 \times \vec{v}_2^\pm) \right|}{\|\vec{v}_1 \times \vec{v}_2^\pm\|}. \quad (10)$$

Write $\vec{v}_1 \times \vec{v}_2^\pm = \langle N_x^\pm, N_y^\pm, N_z^\pm \rangle$. From (5)–(7),

$$N_x^\pm = (\sin \delta (1 - \cos \Delta) \pm \cos \delta \sin \varphi \sin \Delta) \cos \delta \cos \varphi,$$

$$N_y^\pm = (\cos \delta \sin \varphi (1 \mp \cos \Delta) - \sin \delta \sin \Delta) \cos \delta \cos \varphi,$$

$$N_z^\pm = (-1 \pm 1) \sin \delta \cos \delta \sin \varphi \cos \Delta + \sin \Delta \left(\sin^2 \delta \pm \cos^2 \delta \sin^2 \varphi \right).$$

The line l_2^+ . Here $\vec{p}_2^+ - \vec{p}_1 = \cos \varphi \langle \cos \Delta - 1, \sin \Delta, 0 \rangle$, and expanding,

$$(\vec{p}_2^+ - \vec{p}_1) \cdot (\vec{v}_1 \times \vec{v}_2^+) = -2 \cos^2 \varphi \sin \delta \cos \delta (1 - \cos \Delta),$$

$$\|\vec{v}_1 \times \vec{v}_2^+\|^2 = (\sin^2 \delta + \cos^2 \delta \sin^2 \varphi) \left(2(1 - \cos \Delta) \cos^2 \delta \cos^2 \varphi + \sin^2 \Delta (\sin^2 \delta + \cos^2 \delta \sin^2 \varphi) \right).$$

For $\Delta \in (0, 2\pi)$ the norm is positive, since its second factor is at least $2(1 - \cos \Delta) \cos^2 \delta \cos^2 \varphi > 0$, so (10) applies. For $\Delta = 0$ the lines coincide and both sides of (8) vanish. Substituting, dividing numerator and denominator by $\cos^2 \delta$, and using $1 - \cos^2 \Delta = (1 - \cos \Delta)(1 + \cos \Delta)$, $1 - \cos \Delta = 2 \sin^2 \frac{\Delta}{2}$ and $\cos^2 \varphi + \tan^2 \delta + \sin^2 \varphi = 1 + \tan^2 \delta$ gives (8). The bracket in its denominator is at least $1 + \tan^2 \delta - (\tan^2 \delta + \sin^2 \varphi) = \cos^2 \varphi > 0$.

The line l_2^- . Here $\vec{p}_2^- - \vec{p}_1^- = \langle \cos \varphi (\cos \Delta - 1), \cos \varphi \sin \Delta, -2 \sin \varphi \rangle$. With the abbreviations

$$A = \sin \varphi \cos \delta \cos \frac{\Delta}{2} - \sin \delta \sin \frac{\Delta}{2}, \quad B = \sin \delta \sin \varphi \cos \frac{\Delta}{2} + \cos \delta \sin \frac{\Delta}{2}, \quad C = \sin \varphi \cos \delta \sin \frac{\Delta}{2} + \sin \delta \cos \frac{\Delta}{2}, \quad (11)$$

a direct expansion, using the half-angle formulas, gives

$$(\vec{p}_2^- - \vec{p}_1^-) \cdot (\vec{v}_1^- \times \vec{v}_2^-) = 4AB, \quad N_z^- = -2AC, \quad (N_x^-)^2 + (N_y^-)^2 = 4 \cos^2 \delta \cos^2 \varphi A^2, \quad (12)$$

so $\|\vec{v}_1^- \times \vec{v}_2^-\|^2 = 4A^2 [\cos^2 \delta \cos^2 \varphi + C^2]$. The mixed terms in $A^2 + C^2$ cancel, whence

$$A^2 + C^2 = \sin^2 \varphi \cos^2 \delta + \sin^2 \delta, \quad \text{i.e.} \quad \cos^2 \delta \cos^2 \varphi + A^2 + C^2 = 1. \quad (13)$$

In particular $\cos^2 \delta \cos^2 \varphi + C^2 \geq \cos^2 \delta \cos^2 \varphi > 0$, and the two direction vectors are parallel if and only if $A = 0$.

If $A \neq 0$, then (10) and (12) give

$$\text{dist}^2(l_1, l_2^-) = \frac{4B^2}{\cos^2 \delta \cos^2 \varphi + C^2}. \quad (14)$$

For $\Delta = \pi$ (where $A = -\sin \delta \neq 0$) we have $B = \cos \delta$, $C = \sin \varphi \cos \delta$, and (14) equals 4, proving the last claim. For $\Delta \neq \pi$, dividing numerator and denominator of (14) by $\cos^2 \delta \cos^2 \frac{\Delta}{2} > 0$ and using $\tan^2 \delta = \cos^2 \varphi \tan^2 \delta + \sin^2 \varphi \tan^2 \delta$ turns the denominator into $\cos^2 \varphi (1 + \tan^2 \delta) + (\sin \varphi \tan \delta + \tan \frac{\Delta}{2})^2$, which is (9).

If $A = 0$, the parallel unit vectors $\vec{v}_1^-, \vec{v}_2^-$ have the same positive third coordinate $\cos \delta \cos \varphi$, so they are equal. Moreover $(\vec{p}_2^- - \vec{p}_1^-) \cdot \vec{v}_1^- = -2 \cos \varphi \cos \frac{\Delta}{2} A = 0$, so the distance between the two parallel lines is $|\vec{p}_2^- - \vec{p}_1^-|$, whose square $2 \cos^2 \varphi (1 - \cos \Delta) + 4 \sin^2 \varphi = 4(A^2 + B^2) = 4B^2$ agrees with (14), since (13) with $A = 0$ gives $\cos^2 \delta \cos^2 \varphi + C^2 = 1$. Finally $A = 0$ forces $\tan \frac{\Delta}{2} = \sin \varphi \cot \delta > 0$, so $\Delta \neq \pi$ and the reduction to (9) applies verbatim. $\square$

3.2. The formulas in S and T . Throughout the pinwheel construction we abbreviate

$$S = \sin \varphi \in (0, 1), \quad T = \tan \delta > 0, \quad c_0 = (1 - S^2)(1 + T^2).$$

By Lemma 3.2, every distance in the pinwheel is an algebraic function of S and T . In Section 4 we choose S and T rational, so that the inequalities we need become comparisons of rational numbers.

With $\cos^4 \varphi = (1 - S^2)^2$, (8) reads

$$D_{\text{same}}(\Delta)^2 := \text{dist}^2(l_1, l_2^+) = \frac{4 \sin^2 \frac{\Delta}{2} (1 - S^2)^2 T^2}{(S^2 + T^2)(1 + T^2 - \sin^2 \frac{\Delta}{2} (S^2 + T^2))}, \quad (15)$$

and the bracket in the denominator is at least $1 - S^2 > 0$, as shown in the proof of Lemma 3.2. Writing $w = \tan \frac{\Delta}{2} + ST$, the identity $\cos^2 \varphi (1 + \tan^2 \delta) = c_0$ turns (9) into

$$D_{\text{cross}}(\Delta) := \text{dist}(l_1, l_2^-) = \frac{2|w|}{\sqrt{w^2 + c_0}}, \quad w = \tan \frac{\Delta}{2} + ST. \quad (16)$$

We define $D_{\text{cross}}(\pi) = 2$ to be consistent with the limit as $|w| \rightarrow \infty$. Both formulas are 2π -periodic in Δ . For (16) we take $\Delta \in (-\pi, \pi]$, on which $\Delta \mapsto \tan \frac{\Delta}{2}$ is an increasing bijection of $(-\pi, \pi)$ onto $\mathbb{R}$.

The two formulas cover all pairs of pinwheel lines. A rotation about the z -axis carries two up lines with longitudes θ_1, θ_2 to the pair (l_1, l_2^+) with $\Delta = \theta_2 - \theta_1$, and carries an up line and a down line to the pair (l_1, l_2^-) with Δ equal to the longitude of the down line minus that of the up line. The reflection $(x, y, z) \mapsto (x, y, -z)$ carries two down lines to two lines of up type with the same longitudes and tilt $-\delta$, and (15) depends on δ only through T^2 . Hence two lines of the same group with longitudes θ_1, θ_2 are at distance $D_{\text{same}}(\theta_2 - \theta_1)$, and an up line and a down line whose longitudes differ by Δ are at distance $D_{\text{cross}}(\Delta)$.

Lemma 3.3. *As a function of $x = \sin^2 \frac{\Delta}{2} \in (0, 1]$, the right-hand side of (15) is strictly increasing. Consequently, within either group of $P(N; \varphi, \delta, \kappa)$ the minimum distance is attained by adjacent lines and equals $D_{\text{same}}(\gamma)$.*

Proof. The right-hand side is $\frac{Ax}{B-Cx}$ with $A = 4(1 - S^2)^2 T^2 > 0$, $B = (S^2 + T^2)(1 + T^2)$, $C = (S^2 + T^2)^2$, and $B - Cx \geq (S^2 + T^2)(1 - S^2) > 0$ on $[0, 1]$ by the bound recorded after (15). Such a function is strictly increasing. Within a group the longitude differences are $\Delta = j\gamma$, $1 \leq j \leq k - 1$, and $\frac{j\gamma}{2} \in [\frac{\gamma}{2}, \pi - \frac{\gamma}{2}]$, so $\sin^2 \frac{j\gamma}{2} \geq \sin^2 \frac{\gamma}{2}$, with equality for $j \in \{1, k - 1\}$. $\square$

3.3. Criterion in S and T . By Definition 3.1, the longitude of D_i minus the longitude of U_j is

$$\Delta_{ij} = \frac{\gamma}{2} + 2\kappa + (i - j)\gamma \pmod{2\pi},$$

so the cross-group longitude differences form the set

$$\Lambda = \Lambda(\kappa) = \left\{ \frac{\gamma}{2} + 2\kappa + j\gamma \pmod{2\pi} : j = 0, 1, \dots, k - 1 \right\}, \quad (17)$$

an arithmetic progression of step γ on the circle whose offset 2κ we are free to choose.

Lemma 3.4. *Let $\tau^* = 2 \sin \theta$ with $\theta \in (0, \frac{\pi}{2})$, and set*

$$u^* = \sqrt{c_0} \frac{\tau^*}{\sqrt{4 - \tau^{*2}}} = \sqrt{c_0} \tan \theta. \quad (18)$$

Then for $\Delta \in (-\pi, \pi]$,

$$D_{\text{cross}}(\Delta) < \tau^* \iff \Delta \in J := \left(2 \arctan(-ST - u^*), 2 \arctan(-ST + u^*) \right),$$

an open interval of length

$$|J| = 2 \left[\arctan(a + u^*) - \arctan(a - u^*) \right], \quad a = ST > 0.$$

Moreover, if $|J| \leq \gamma$, then for

$$\kappa = \kappa^* := -\frac{1}{2} \left[\arctan(ST - u^*) + \arctan(ST + u^*) \right] \quad (19)$$

the lattice (17) is disjoint from J , and hence every cross-group distance in $P(N; \varphi, \delta, \kappa^)$ is at least τ^* .*

Proof. The function $t \mapsto \frac{2t}{\sqrt{t^2 + c_0}}$ is strictly increasing on $[0, \infty)$, and u^* of (18) is the value at which it equals τ^* .

Hence $D_{\text{cross}}(\Delta) < \tau^*$ if and only if $|w| < u^*$. Now $|w| < u^*$ means $\tan \frac{\Delta}{2} \in (-ST - u^*, -ST + u^*)$, and as $\Delta \mapsto \tan \frac{\Delta}{2}$ is an increasing bijection from $(-\pi, \pi)$ onto $\mathbb{R}$ (and $D_{\text{cross}}(\pi) = 2 \geq \tau^*$), the failure set is exactly the interval J . By the oddness of $\arctan$,

$$|J| = 2 \left[\arctan(u^* - ST) + \arctan(u^* + ST) \right] = 2 \left[\arctan(a + u^*) - \arctan(a - u^*) \right], \quad a = ST.$$

The interval J is centered at $c = \arctan(-ST - u^*) + \arctan(-ST + u^*) = 2\kappa^*$. With $\kappa = \kappa^*$ the lattice (17) is $\{c + \frac{\gamma}{2} + j\gamma\}$, whose distance (on the circle) from the center c is at least $\frac{\gamma}{2} \geq \frac{|J|}{2}$. Since J is open, no lattice point lies in J . $\square$

Fix an even $N \geq 6$, parameters $S \in (0, 1)$, $T > 0$, and a target $\tau^* = 2 \sin \theta$ with $\theta \in (0, \frac{\pi}{2})$. By Lemma 3.3, all same-group distances are at least τ^* if and only if $D_{\text{same}}(\frac{4\pi}{N})^2 \geq \tau^{*2}$. Multiplied out, this *same-group inequality* reads

$$\sin^2 \frac{2\pi}{N} (1 - S^2)^2 T^2 \geq \sin^2 \theta \left[(S^2 + T^2)(1 + T^2) - \sin^2 \frac{2\pi}{N} (S^2 + T^2)^2 \right]. \quad (\text{same-group})$$

Lemma 3.4 shows that all cross-group distances are at least τ^* once the phase is chosen by (19), provided the interval J on which $D_{\text{cross}} < \tau^*$ is no longer than $\frac{4\pi}{N}$. This is the *cross-group inequality*

$$\arctan(a + \sqrt{c_0} \tan \theta) - \arctan(a - \sqrt{c_0} \tan \theta) \leq \frac{2\pi}{N}. \quad (\text{cross-group})$$

Theorem 3.5. *Let $N = 2k \geq 6$ be even, $S \in (0, 1)$, $T > 0$, and $\tau^* = 2 \sin \theta$ with $\theta \in (0, \frac{\pi}{2})$. If (same-group) and (cross-group) hold, then the pinwheel $P(N; \varphi, \delta, \kappa^*)$ with $\sin \varphi = S$, $\tan \delta = T$ and κ^* as in (19) consists of N lines tangent to $\mathbb{S}^2$ with all pairwise distances at least τ^* . Consequently, by Lemma 2.1, at least N pairwise non-overlapping cylinders of radius $\frac{\sin \theta}{1 - \sin \theta}$ touch the unit ball.*

Proof. Same-group pairs: by Lemma 3.3 the minimum is $D_{\text{same}}(\gamma)$ with $\gamma = \frac{4\pi}{N}$, and (same-group) is exactly $D_{\text{same}}(\gamma)^2 \geq \tau^{*2}$ after multiplying out the (positive) denominator of (15) at $\sin^2 \frac{\gamma}{2} = \sin^2 \frac{2\pi}{N}$. Cross-group pairs: (cross-group) is $|J| \leq \gamma$, so Lemma 3.4 applies. The cylinder statement follows since $\tau^* = 2 \sin \theta$ corresponds under (3) to $\omega = \frac{\sin \theta}{1 - \sin \theta}$. $\square$

Remark 3.6. Taking $\theta = \frac{\pi}{N-1}$, the cylinders in Theorem 3.5 have radius exactly ω_{N-1} , and $N = (N-1) + 1$ of them touch B . Taking $\theta = \frac{\pi}{N-2}$ gives $N = (N-2) + 2$ cylinders of radius exactly ω_{N-2} . These are the two cases used in Section 4.

4. PROOF OF THEOREM 1.12

4.1. **The frozen parameters.** From now on $S = \frac{9}{20}$, $T = \frac{31}{64}$, so that all derived constants are rational:

$$a = ST = \frac{279}{1280}, \quad a^2 = \frac{77841}{1638400}, \quad c_0 = (1 - S^2)(1 + T^2) = \frac{1613183}{1638400}.$$

We will need

$$2\Phi > \frac{21}{20} \quad \text{and} \quad \frac{21}{20}\sqrt{c_0} < 1 + a^2, \quad (20)$$

$$\text{where } \Phi \text{ is defined by } \Phi^2 = \frac{(1 - S^2)^2 T^2}{(S^2 + T^2)(1 + T^2)} = \frac{1564677136}{5658909425}.$$

After squaring, both inequalities in (20) are comparisons of rational numbers, checked by the verifier (Table 1).

4.2. **Elementary estimates.** Appendix A collects, with one-line proofs, the only analytic inequalities we use:

$$(E1) \quad x - \frac{x^3}{6} \leq \sin x \leq x \text{ for } x \geq 0;$$

$$(E2) \quad \tan y \leq y + y^3 \text{ for } y \in (0, 1];$$

$$(E3) \quad \arctan p - \arctan q = \arctan \frac{p-q}{1+pq} \text{ whenever } pq > -1, \text{ and } 0 \leq \arctan z \leq z \text{ for } z \geq 0;$$

$$(E4) \quad z \mapsto \frac{z}{1+a^2-z^2} \text{ is strictly increasing on } [0, \sqrt{1+a^2});$$

$$(E5) \quad \frac{1}{\sqrt{1-z^2}} \leq 1 + z^2 \text{ for } 0 \leq z \leq \frac{3}{4}.$$

4.3. **The two tail lemmas.** Fix $s \in \{1, 2\}$ and put $\theta = \theta(N) = \frac{\pi}{N-s}$ in (same-group)–(cross-group).

Lemma 4.1. *Let $N \geq 2s + 4$ be even. If*

$$2\left(1 - \frac{s}{N}\right)\left(1 - \frac{2\pi^2}{3N^2}\right) \geq \frac{1}{\Phi}, \quad (21)$$

then (same-group) holds with $\theta = \frac{\pi}{N-s}$. The left-hand side of (21) is strictly increasing in N , and (21) holds for $s = 1$, $N = 32$ and for $s = 2$, $N = 48$. Hence (same-group) holds for all even $N \geq 32$ (respectively $N \geq 48$).

Proof. Discarding the subtracted term on the right of (same-group) shows that $\sin \frac{2\pi}{N} \Phi \geq \sin \theta$ is sufficient. By (E1), $\sin \frac{2\pi}{N} \geq \frac{2\pi}{N} \left(1 - \frac{2\pi^2}{3N^2}\right)$ and $\sin \theta \leq \theta = \frac{\pi}{N-s}$, so (21) suffices. The left side of (21) is a product of two positive factors, each strictly increasing in N . For the two threshold cases, (20) gives $\frac{1}{\Phi} < \frac{40}{21} < 1.905$. Using $\pi^2 < 10$, the left-hand side of (21) is at least $\frac{31}{16} \cdot \frac{763}{768} > 1.92$ at $(s, N) = (1, 32)$ and at least $\frac{23}{12} \cdot \frac{1723}{1728} > 1.91$ at $(s, N) = (2, 48)$. $\square$

Lemma 4.2. *Let $N \geq 2s + 4$ be even, $\theta = \frac{\pi}{N-s} \leq 1$, and set $\bar{u} = \sqrt{c_0}(\theta + \theta^3)$. If $\bar{u}^2 < 1 + a^2$ and*

$$N\bar{u} \leq \pi(1 + a^2 - \bar{u}^2), \quad (22)$$

then (cross-group) holds with $\theta = \frac{\pi}{N-s}$. The left-hand side of (22) is strictly decreasing in N and the right-hand side is increasing, so (22), once true, remains true for all larger N . It holds for $s = 1$, $N = 32$ and for $s = 2$, $N = 48$, hence for all even $N \geq 32$ (respectively $N \geq 48$).

Proof. By (E2), $u^* = \sqrt{c_0} \tan \theta \leq \bar{u}$, so $u^{*2} \leq \bar{u}^2 < 1 + a^2$. By (E3) (the product $(a + u^*)(a - u^*) = a^2 - u^{*2} > -1$),

$$\arctan(a + u^*) - \arctan(a - u^*) = \arctan \frac{2u^*}{1 + a^2 - u^{*2}} \leq \frac{2u^*}{1 + a^2 - u^{*2}} \leq \frac{2\bar{u}}{1 + a^2 - \bar{u}^2} \leq \frac{2\pi}{N},$$

using (E4) and (22). For the monotonicity, $N\bar{u} = \pi\sqrt{c_0} \frac{N}{N-s} (1 + \theta^2)$ is a product of positive strictly decreasing factors of N , while $\bar{u}$ itself decreases, so $\pi(1 + a^2 - \bar{u}^2)$ increases. The two threshold cases are in Table 1. $\square$

4.4. The computer-checked inequalities. Table 1 lists every numerical inequality that the pinwheel construction relies on. In each of them, the ingredients π , $\sin$, $\tan$, $\arctan$ and $\sqrt{c_0}$ are replaced by rational bounds. Each bound is chosen on the side that makes the resulting inequality imply the original one. What remains is a comparison of two explicit rational numbers. The standalone verifier [11], invoked as `cylie -i constructions`, performs all of them in exact arithmetic, with 512-bit overflow-checked integers and no floating point. The same run performs the checks (W1)–(W3) of Section 6 and ends with the line `ALL CONSTRUCTIONS VERIFIED`.

inequality	checked at	used in
(same-group) and (cross-group) with $\theta = \frac{\pi}{N-1}$	every even N with $20 \leq N \leq 30$	Theorem 1.12, $20 \leq N \leq 30$
(same-group) and (cross-group) with $\theta = \frac{\pi}{N-2}$	every even N with $40 \leq N \leq 46$	Theorem 1.12, $40 \leq N \leq 46$
(22)	$(s, N) = (1, 32)$ and $(2, 48)$	Lemma 4.2
(20)	independent of N	Lemma 4.1, Corollary 1.8

TABLE 1. The numerical inequalities of the pinwheel construction, all checked by the verifier.

4.5. Proof of Theorem 1.12. Let $n \geq 19$ be odd and $N = n + 1$, so N is even, $N \geq 20$, and $\theta = \frac{\pi}{n} = \frac{\pi}{N-1}$. If $20 \leq N \leq 30$, inequalities (same-group) and (cross-group) hold by Table 1. If $N \geq 32$, the inequalities hold by Lemmas 4.1 and 4.2. By Theorem 3.5 and Remark 3.6, $N = n + 1$ pairwise non-overlapping cylinders of radius ω_n touch B , so $\mathcal{N}(\omega_n) \geq n + 1$.

Let $n \geq 38$ be even and $N = n + 2$, so N is even, $N \geq 40$, and $\theta = \frac{\pi}{n} = \frac{\pi}{N-2}$. If $40 \leq N \leq 46$, use Table 1. If $N \geq 48$, use Lemmas 4.1 and 4.2. As before, $\mathcal{N}(\omega_n) \geq n + 2$. $\square$

5. THIN CYLINDERS: PROOF OF COROLLARY 1.8

We apply Theorem 3.5 with the target

$$\sin \theta_N = \frac{21\pi}{20N},$$

which is admissible for all $N \geq 4$ since $\frac{21\pi}{20N} < 1$. Only large N matters here, so no finite table is needed.

Lemma 5.1. *There is an even N_0 such that, for every even $N \geq N_0$, inequalities (same-group) and (cross-group) hold with $\theta = \theta_N = \arcsin \frac{21\pi}{20N}$.*

Proof. Condition (same-group). As in Lemma 4.1, it suffices to prove that $\sin \frac{2\pi}{N} \Phi \geq \sin \theta_N = \frac{21\pi}{20N}$, and by (E1) it suffices to prove that $1 - \frac{2\pi^2}{3N^2} \geq \frac{21}{40\Phi}$. The left side increases to 1, and $\frac{21}{40\Phi} < 1$ is the first inequality of (20). So the condition holds for all large N .

Condition (cross-group). Write $z = \frac{21\pi}{20N}$, which is below $\frac{3}{4}$ for $N \geq 5$. By (E5), $\tan \theta_N = \frac{z}{\sqrt{1-z^2}} \leq z(1+z^2)$, so $u^* = \sqrt{c_0} \tan \theta_N \leq \bar{u} := \sqrt{c_0} z(1+z^2)$, and as in Lemma 4.2 it suffices to prove that $\bar{u}^2 < 1 + a^2$ and $N\bar{u} \leq \pi(1 + a^2 - \bar{u}^2)$. As $N \rightarrow \infty$, $\bar{u} \rightarrow 0$, $N\bar{u} = \frac{21\pi}{20} \sqrt{c_0} (1 + z^2)$ decreases to $\frac{21\pi}{20} \sqrt{c_0}$, and the right side increases to $\pi(1 + a^2)$. Since $\frac{21}{20} \sqrt{c_0} < 1 + a^2$ is the second inequality of (20), both conditions hold for all large N . $\square$

Proof of Corollary 1.8. For even $N \geq N_0$, Lemma 5.1 and Theorem 3.5 produce N pairwise non-overlapping cylinders of radius

$$\omega_N^* = \frac{\sin \theta_N}{1 - \sin \theta_N} > \sin \theta_N = \frac{21\pi}{20N}$$

touching B . By Lemma 1.5, $\mathcal{N}(\omega) \geq N$ for every $\omega \leq \omega_N^*$. Given $0 < \omega < \omega_{N_0}^*$, let $N(\omega)$ be the largest even $N \geq N_0$ with $\omega_N^* \geq \omega$. Then $\omega > \omega_{N(\omega)+2}^* > \frac{21\pi}{20(N(\omega)+2)}$ and $\mathcal{N}(\omega) \geq N(\omega)$, hence

$$\omega \mathcal{N}(\omega) > \frac{21\pi}{20} \cdot \frac{N(\omega)}{N(\omega) + 2}.$$

As $\omega \rightarrow 0^+$ we have $N(\omega) \rightarrow \infty$ (since $\omega_N^* \rightarrow 0$), and the right side tends to $\frac{21\pi}{20}$. $\square$

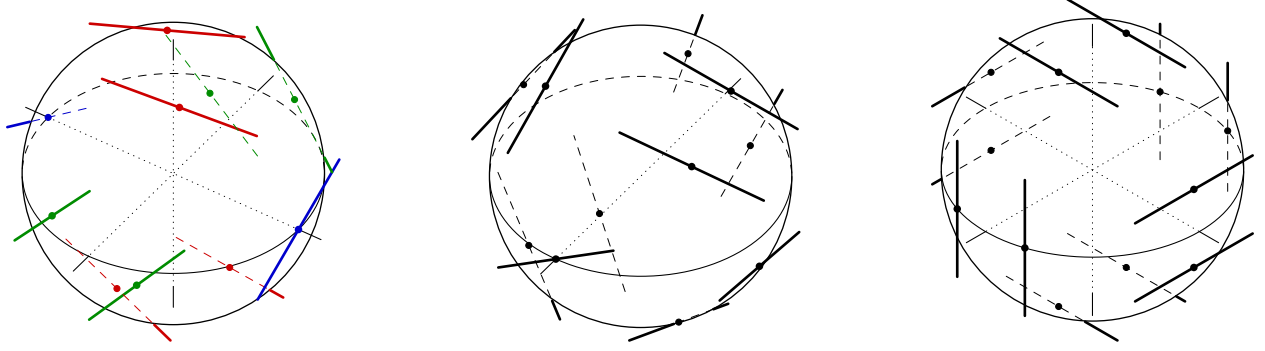


FIGURE 4. The configurations that show $(9, \omega_9)$, $(10, \omega_{10})$ and $(11, \omega_{11}) \notin \mathcal{P}_{\text{cyl}}$. Their symmetries and a few other features are described in Section 14.1.

6. THE WINDOW: EXPLICIT CONFIGURATIONS FOR $9 \leq n \leq 36$

For each n in the window left open by Theorems 1.6 and 1.12, except $n = 8$, we give $n + 1$ tangent lines with rational data. Their pairwise distances are computed exactly and compared with a rational upper bound for $2 \sin \frac{\pi}{n}$. The pinwheel is not used in this section.

Theorem 6.1. *For every $n \in W := \{9, 10, \dots, 18\} \cup \{20, 22, \dots, 36\}$,*

$$N(\omega_n) \geq n + 1, \quad \text{i.e.} \quad (n, \omega_n) \notin \mathcal{P}_{\text{cyl}}.$$

Proof of Theorem 1.7. Theorem 6.1 covers W , and Theorem 1.12 covers every odd $n \geq 19$ and every even $n \geq 38$. The union is $\{n \geq 9\}$. $\square$

Together with Theorem 1.6, the insertion question (Question 1.1) is decided for every $n \geq 3$ except $n = 8$: the ring of n is count-optimal at its own radius for $3 \leq n \leq 7$ and is not for any $n \geq 9$.

6.1. Rational tangent lines. The witnesses are lines tangent to $\mathbb{S}^2$ at rational points, with integer directions. Normalizing the direction is unnecessary, because the distance formula of Lemma 2.2 is homogeneous of degree zero in each direction vector.

Definition 6.2. A *rational tangent line* is a pair (P, V) of integer vectors $P = (A, B, C, E) \in \mathbb{Z}^4$ and $V = (V_0, V_1, V_2) \in \mathbb{Z}^3 \setminus \{0\}$ with

$$(W1) \quad A^2 + B^2 + C^2 = E^2, \quad E > 0, \quad (W2) \quad AV_0 + BV_1 + CV_2 = 0, \quad V \neq 0. \quad (23)$$

It denotes the line $\ell(P, V) = \{p + tV : t \in \mathbb{R}\}$ with $p = \frac{1}{E}(A, B, C)$, which by (W1) and (W2) is tangent to the unit sphere at p .

The following result is an immediate consequence of Lemma 2.2.

Lemma 6.3. *Let $\ell = \ell(P, V)$ and $\ell' = \ell(P', V')$ be rational tangent lines with tangency points p, p' . Define $c = V \times V'$ and $\Delta = p' - p$. Then $\text{dist}(\ell, \ell')^2$ is the rational number*

$$\frac{(c \cdot \Delta)^2}{|c|^2} \quad \text{if } c \neq 0, \quad \left| \Delta - \frac{\Delta \cdot V}{|V|^2} V \right|^2 \quad \text{if } c = 0. \quad (24)$$

The threshold $2 \sin \frac{\pi}{n}$ is irrational. We replace it by a slightly larger rational number.

Lemma 6.4. *For every integer $n \geq 1$ let $Y_n = \frac{710}{113n}$ and*

$$r_n = Y_n^2 - \frac{Y_n^4}{12} + \frac{Y_n^6}{360} \in \mathbb{Q}.$$

Then $4 \sin^2 \frac{\pi}{n} \leq r_n$.

Proof. For $y \geq 0$,

$$\cos y \geq 1 - \frac{y^2}{2} + \frac{y^4}{24} - \frac{y^6}{720}. \quad (25)$$

Indeed, the difference of the two sides vanishes at 0, its derivative $-\sin y + y - \frac{y^3}{6} + \frac{y^5}{120}$ vanishes at 0, the derivative of that, $-\cos y + 1 - \frac{y^2}{2} + \frac{y^4}{24}$, vanishes at 0, and its derivative $\sin y - y + \frac{y^3}{6}$ is nonnegative by (E1); integrating three times from 0 gives (25). Hence $4 \sin^2 \frac{\pi}{n} = 2(1 - \cos \frac{2\pi}{n}) \leq \rho(2\pi/n)$ with $\rho(y) = y^2 - \frac{y^4}{12} + \frac{y^6}{360}$. Now $\rho'(y) = \frac{y}{60}(y^4 - 20y^2 + 120) > 0$ for $y > 0$ (the quadratic in y^2 has negative discriminant), so ρ is increasing on $[0, \infty)$, and since $\pi < \frac{355}{113}$ we have $\frac{2\pi}{n} < Y_n$ and $\rho(2\pi/n) \leq \rho(Y_n) = r_n$. $\square$

6.2. The deposited configurations and their verification. For each $N \in \{10, 11, 12, \dots, 21\} \cup \{23, 25, \dots, 37\}$ a file

`cyl.window_N{N}.txt`

contains a configuration that, for $n = N - 1$, proves $(n, \omega_n) \notin \mathcal{P}_{\text{cyl}}$. The file contains N rows of seven integers $A B C E V_0 V_1 V_2$. Each row corresponds to one rational tangent line. All integers have at most 13 decimal digits. The verifier (Section 4.4) performs the following checks:

- (W1) for every row, $A^2 + B^2 + C^2 = E^2$ and $E > 0$;
- (W2) for every row, $AV_0 + BV_1 + CV_2 = 0$ and $(V_0, V_1, V_2) \neq (0, 0, 0)$;
- (W3) for every pair of rows, the rational squared distance of Lemma 6.3 is at least r_n .

Proof of Theorem 6.1. Let $n \in W$ and $N = n + 1$. The deposited file with N rows passes (W1)–(W3). By (W1) and (W2) the rows are N lines tangent to $\mathbb{S}^2$. By Lemma 6.3 and (W3) their pairwise squared distances are at least r_n , and by Lemma 6.4 $r_n \geq 4 \sin^2 \frac{\pi}{n} = \tau(\omega_n)^2$ (Remark 1.11). So N tangent lines have pairwise distances at least $\tau(\omega_n)$, and Lemma 2.1 gives $N = n + 1$ non-overlapping cylinders of radius ω_n touching B . $\square$

7. THE TWO-POINT BOUND

The bound we use is the two-point bound for packing problems on a compact group; for the rotation group it was first developed by de Oliveira Filho, Spomer and Vallentin [23], within the general setting of Bachoc and Vallentin [1]. We state and prove it in the form we need.

Throughout this section, $\tau \in (0, 2)$ is fixed, $L \in \mathbb{N}$, and, for $\ell = 1, \dots, L$, we fix

- (i) a symmetric positive-definite matrix $D_\ell \in \mathbb{R}^{n_\ell \times n_\ell}$, and
- (ii) a map $T^\ell : \text{SO}(3) \rightarrow \mathbb{R}^{n_\ell \times n_\ell}$ satisfying, for all $g_1, g_2, g \in \text{SO}(3)$,

$$T^\ell(g_1 g_2) = T^\ell(g_1) T^\ell(g_2) \quad \text{and} \quad T^\ell(g)^\top D_\ell T^\ell(g) = D_\ell.$$

Lemma 7.1. *Under (i)–(ii), for every $g \in \text{SO}(3)$: $\det T^\ell(g) = \pm 1$; $T^\ell(e) = I$; and*

$$T^\ell(g^{-1}) = T^\ell(g)^{-1} = D_\ell^{-1} T^\ell(g)^\top D_\ell.$$

Proof. Taking determinants in $T^\top D T = D$ gives $(\det T)^2 \det D = \det D$, and $\det D > 0$, so $\det T = \pm 1$. In particular every $T^\ell(g)$ is invertible. Then $T(e) = T(e \cdot e) = T(e)^2$ with $T(e)$ invertible forces $T(e) = I$, and $T(g)T(g^{-1}) = T(e) = I$ gives $T(g^{-1}) = T(g)^{-1}$. Finally, multiplying $T^\top D T = D$ on the left by D^{-1} and on the right by T^{-1} gives $D^{-1} T^\top D = T^{-1}$. $\square$

Definition 7.2. For a sequence $\vec{C} = (C_\ell)_{\ell=1}^L$ of symmetric positive-semidefinite matrices $C_\ell \in \mathbb{R}^{n_\ell \times n_\ell}$, we define its supremum over the safe zone as

$$S(\vec{C}) = \sup_{g \in \mathcal{D}(\tau)} \sum_{\ell=1}^L \text{tr}(D_\ell^{-1} C_\ell T^\ell(g)).$$

Remark 7.3. When $D_\ell = \text{diag}(w_1^\ell, \dots, w_{n_\ell}^\ell)$ is diagonal and C_ℓ is symmetric,

$$\text{tr}(D_\ell^{-1} C_\ell T^\ell(g)) = \sum_{i,j} \frac{(C_\ell)_{ij}}{w_j^\ell} T^\ell(g)_{ij},$$

the sum of the entries of the Schur product of C_ℓ and $T^\ell(g)$, with the j -th column weighted by $1/w_j^\ell$. In particular $\text{tr}(D_\ell^{-1} C_\ell T^\ell(e)) = \text{tr}(D_\ell^{-1} C_\ell) = \sum_i (C_\ell)_{ii} / w_i^\ell$.

Theorem 7.4. Let $\vec{C} = (C_\ell)_{\ell=1}^L$ be a sequence of symmetric positive-semidefinite matrices $C_\ell \in \mathbb{R}^{n_\ell \times n_\ell}$. If $S(\vec{C}) < 0$ and M is any real number with $S(\vec{C}) \leq M < 0$, then

$$N_{\max}(\tau) \leq 1 + \left\lceil \frac{\sum_{\ell=1}^L \text{tr}(D_\ell^{-1} C_\ell)}{-M} \right\rceil. \quad (26)$$

Proof. Let $g_1, \dots, g_N$ be distinct elements of a non-overlapping set (Definition 2.4), and put $S_\ell = \sum_{i=1}^N T^\ell(g_i)$. By Lemma 7.1,

$$T^\ell(g_i^{-1} g_j) = T^\ell(g_i)^{-1} T^\ell(g_j) = D_\ell^{-1} T^\ell(g_i)^\top D_\ell T^\ell(g_j),$$

so that, summing over all ordered pairs and using linearity and cyclicity of the trace,

$$\begin{aligned} \sum_{i=1}^N \sum_{j=1}^N \text{tr}(D_\ell^{-1} C_\ell T^\ell(g_i^{-1} g_j)) &= \sum_{i,j} \text{tr}(D_\ell^{-1} C_\ell D_\ell^{-1} T^\ell(g_i)^\top D_\ell T^\ell(g_j)) \\ &= \text{tr}((D_\ell^{-1} C_\ell D_\ell^{-1})(S_\ell^\top D_\ell S_\ell)) \geq 0, \end{aligned}$$

since $D_\ell^{-1} C_\ell D_\ell^{-1}$ and $S_\ell^\top D_\ell S_\ell$ are congruent to C_ℓ and D_ℓ , hence symmetric positive semidefinite, and for such P, Q we have $\text{tr}(PQ) = \text{tr}(\sqrt{Q} P \sqrt{Q}) \geq 0$.

Summing over ℓ and separating the N diagonal pairs $i = j$ (where $T^\ell(e) = I$ gives the value $\sum_\ell \text{tr}(D_\ell^{-1} C_\ell)$) from the $N(N-1)$ off-diagonal pairs (where $g_i^{-1} g_j \in \mathcal{D}$ by Definition 2.4, so each term is at most $S(\vec{C}) \leq M$),

$$0 \leq N \sum_{\ell=1}^L \text{tr}(D_\ell^{-1} C_\ell) + N(N-1)M.$$

Since $M < 0$, dividing by N and rearranging gives $(N-1)(-M) \leq \sum_\ell \text{tr}(D_\ell^{-1} C_\ell)$, which implies (26). $\square$

8. STRATEGY FOR THE PROOF

Theorem 7.4 turns an upper bound on $N_{\max}(\tau)$ into a question about finitely many objects: matrices D_ℓ, C_ℓ and maps T^ℓ satisfying the two identities of Section 7, together with a number $M < 0$ that dominates the trace sum on the safe zone. A *certificate* is an explicit choice of these objects for a given rational τ^2 . The proof of $(N, \omega) \in \mathcal{P}_{\text{cyl}}$ then consists of exhibiting a certificate and verifying its properties.

The difficulty is that the properties, as stated, quantify over $\text{SO}(3)$. Two ideas make them finitely checkable.

First, we represent $\text{SO}(3)$ by unit quaternions (Section 9). The maps T^ℓ become matrices of polynomials in four variables $q_0, \dots, q_3$ with rational coefficients, the two identities become polynomial identities, and the safe zone $\mathcal{D}(\tau)$ becomes a semialgebraic set. Polynomial identities are verified by expanding both sides.

Second, the inequality “trace sum $\leq M$ on the safe zone” is proved by a sum-of-squares identity (Section 13): the certificate contains multiplier matrices $\Sigma_0, \dots, \Sigma_3$ such that M minus the trace sum equals a polynomial that is visibly nonnegative on the safe zone. This identity is again verified by expansion, and the nonnegativity of the multipliers by an explicit diagonal-dominance test.

Section 10 collects these requirements into six hypotheses (H1)–(H6) and proves that they imply the bound. Section 11 applies the theorem to the five deposited certificates, and Sections 12 and 13 describe the deposited data and the checks the verifier performs. The matrices D_ℓ and T^ℓ are the same for all certificates and deposited once; τ^2, M, C_ℓ and the multipliers are specific to each certificate. No property of the data is assumed: everything the proof uses is either a rational identity or a rational inequality, and the verifier checks each one in exact integer arithmetic.

9. REDUCTION TO A POLYNOMIAL PROBLEM

We will now find an equivalent problem in which $\text{SO}(3)$ does not appear, and whose only objects are polynomials in the four variables q_0, q_1, q_2 , and q_3 .

9.1. Quaternions and rotations. For $q = (q_0, q_1, q_2, q_3) \in \mathbb{R}^4$ define

$$R(q) = \begin{bmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1 q_2 - q_0 q_3) & 2(q_1 q_3 + q_0 q_2) \\ 2(q_1 q_2 + q_0 q_3) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_2 q_3 - q_0 q_1) \\ 2(q_1 q_3 - q_0 q_2) & 2(q_2 q_3 + q_0 q_1) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{bmatrix},$$

whose entries are integer quadratic forms in q , and write $|q|^2 = q_0^2 + q_1^2 + q_2^2 + q_3^2$. Let $p * q$ denote the Hamilton product,

$$\begin{aligned} (p * q)_0 &= p_0 q_0 - p_1 q_1 - p_2 q_2 - p_3 q_3, \\ (p * q)_1 &= p_0 q_1 + q_0 p_1 + p_2 q_3 - p_3 q_2, \\ (p * q)_2 &= p_0 q_2 + q_0 p_2 + p_3 q_1 - p_1 q_3, \\ (p * q)_3 &= p_0 q_3 + q_0 p_3 + p_1 q_2 - p_2 q_1. \end{aligned}$$

Lemma 9.1. *The following hold as polynomial identities in $\mathbb{Z}[p, q]$:*

- (a) $R(q)^\top R(q) = |q|^4 I$ and $\det R(q) = |q|^6$; in particular $R(q) \in \text{SO}(3)$ whenever $|q| = 1$;
- (b) $R(p * q) = R(p) R(q)$ and $|p * q|^2 = |p|^2 |q|^2$;
- (c) the ten products $q_a q_b$ ($0 \leq a \leq b \leq 3$) are $\mathbb{Z}$ -linear combinations of $|q|^2$ and the entries r_{ij} of $R(q)$; explicitly $4q_0^2 = |q|^2 + r_{11} + r_{22} + r_{33}$, $4q_1^2 = |q|^2 + r_{11} - r_{22} - r_{33}$, $4q_2^2 = |q|^2 - r_{11} + r_{22} - r_{33}$, $4q_3^2 = |q|^2 - r_{11} - r_{22} + r_{33}$, $4q_0 q_1 = r_{32} - r_{23}$, $4q_0 q_2 = r_{13} - r_{31}$, $4q_0 q_3 = r_{21} - r_{12}$, $4q_1 q_2 = r_{12} + r_{21}$, $4q_1 q_3 = r_{13} + r_{31}$, $4q_2 q_3 = r_{23} + r_{32}$.

Proof. Each identity is an equality of polynomials with integer coefficients, of degree at most 6 in q in (a) and (c) and at most 4 in p, q in (b), and is verified by expanding both sides. $\square$

Lemma 9.2.

- (a) If q, q' are unit quaternions with $R(q) = R(q')$, then $q' = \pm q$.
- (b) Every $g \in \text{SO}(3)$ equals $R(q)$ for some unit quaternion q .

Proof. (a) By Lemma 9.1(c) the entries of $R(q)$ together with $|q|^2 = 1$ determine every product $q_a q_b$, i.e. the symmetric rank-one matrix $qq^\top$. If $qq^\top = q'q'^\top$ then, taking any index a with $q_a \neq 0$ (such an index exists because $|q| = 1$), the a -th columns give $q_a q = q'_a q'$ with $q_a^2 = (q'_a)^2$, so $q' = \pm q$.

(b) First, every $g \in \text{SO}(3)$ has an eigenvector corresponding to eigenvalue 1. Indeed, since $\det g = 1$ and $g^\top g = I$,

$$\det(g - I) = \det(g) \det(I - g^\top) = \det((I - g)^\top) = \det(I - g) = (-1)^3 \det(g - I).$$

Therefore, $\det(g - I) = 0$ and there is a unit vector n with $gn = n$. Since g is orthogonal and fixes n , it maps the plane $n^\perp$ to itself, and its restriction there is an orientation-preserving linear isometry of a plane, hence a rotation by some angle θ . Let $n = (n_x, n_y, n_z)$ and set

$$q = \left(\cos \frac{\theta}{2}, n_x \sin \frac{\theta}{2}, n_y \sin \frac{\theta}{2}, n_z \sin \frac{\theta}{2} \right).$$

Clearly, q is a unit quaternion. A direct computation from the definition of $R(q)$ shows $R(q)n = n$ and that $R(q)$ rotates $n^\perp$ by θ with the same orientation. Therefore, $R(q) = g$. $\square$

9.2. The safe zone as a pair of polynomial inequalities. By Lemma 9.2 the map $q \mapsto R(q)$ carries the unit sphere of $\mathbb{R}^4$ onto $\text{SO}(3)$, two-to-one. To transport the safe zone $\mathcal{D}(\tau)$ of Definition 2.3 along this map we first express the condition $\text{dist}(\ell_0, g \cdot \ell_0) \geq \tau$ algebraically in g .

Fix $g \in \text{SO}(3)$ and write

$$u = g \cdot u_0, \quad v = g \cdot v_0, \quad m = u - u_0, \quad \rho = v_0 \times v,$$

so that ℓ_0 passes through u_0 with unit direction v_0 , and $g \cdot \ell_0$ passes through u with unit direction v . The vector ρ is the common normal direction of the two lines whenever they are not parallel, and u and v are the third and the first column of g . Define

$$n = m \cdot \rho, \tag{27}$$

$$N_2 = |\rho|^2, \tag{28}$$

$$U_1^2 = |m|^2 - (m \cdot v_0)^2. \tag{29}$$

Writing $c = v_0 \cdot v$, we have $N_2 = 1 - c^2$, so $N_2 > 0$ exactly when v_0, v are linearly independent, that is, when the two lines are skew or intersecting, and $N_2 = 0$ exactly when they are parallel. The quantity U_1 is the distance from the point u to the line ℓ_0 .

Lemma 9.3. *With the notation above,*

$$\text{dist}(\ell_0, g \cdot \ell_0) = \begin{cases} |n|/\sqrt{N_2}, & N_2 > 0, \\ U_1, & N_2 = 0. \end{cases}$$

Moreover $n = 0$ whenever $N_2 = 0$.

Proof. The last assertion is immediate: $N_2 = 0$ means $\rho = 0$, hence $n = m \cdot \rho = 0$.

Case $N_2 > 0$. A general point of ℓ_0 is $u_0 + tv_0$ and a general point of $g \cdot \ell_0$ is $u + sv$, so a general difference is $m + sv - tv_0$. Since ρ is orthogonal to both v_0 and v ,

$$(m + sv - tv_0) \cdot \rho = m \cdot \rho = n,$$

and the Cauchy–Schwarz inequality gives $|m + sv - tv_0| \geq |n|/|\rho| = |n|/\sqrt{N_2}$ for all s, t . For the reverse inequality, note that $\{v_0, v, \rho\}$ is a basis of $\mathbb{R}^3$ because $N_2 > 0$; writing $m = \alpha v_0 + \beta v + \gamma \rho$ and choosing $t = \alpha$, $s = -\beta$ makes the difference equal to $\gamma \rho$, whose norm is $|\gamma| |\rho| = |m \cdot \rho|/|\rho| = |n|/\sqrt{N_2}$. Hence the infimum is attained and equals $|n|/\sqrt{N_2}$.

Case $N_2 = 0$. Here $v = \pm v_0$, so the two lines are parallel and the distance between them equals the distance from any point of one to the other; taking the point u gives $\text{dist}(\ell_0, g \cdot \ell_0) = U_1$. $\square$

Let us define

$$Q = n^2 - \tau^2 N_2, \tag{30}$$

$$U = U_1^2 - \tau^2. \tag{31}$$

Theorem 9.4. *For every $g \in \text{SO}(3)$ we have $g \in \mathcal{D}(\tau)$ if and only if $Q \geq 0$ and $U \geq 0$. In short, $\mathcal{D}(\tau) = \{Q \geq 0\} \cap \{U \geq 0\}$.*

Proof. Write $\delta = \text{dist}(\ell_0, g \cdot \ell_0)$, so that $g \in \mathcal{D}(\tau)$ means $\delta \geq \tau$. Since u lies on $g \cdot \ell_0$, we have $\delta \leq U_1$ always.

Case $N_2 > 0$. By Lemma 9.3, $\delta^2 N_2 = n^2$, so $Q = N_2(\delta^2 - \tau^2)$ and, as $N_2 > 0$, the sign of Q is the sign of $\delta^2 - \tau^2$; hence $Q \geq 0 \iff \delta \geq \tau$. If $Q \geq 0$ then $U_1 \geq \delta \geq \tau$, so $U \geq 0$ automatically. Therefore, on $\{N_2 > 0\}$, both inequalities hold exactly when $\delta \geq \tau$.

Case $N_2 = 0$. Then $n = 0$ by Lemma 9.3, so $Q = 0$: the first inequality holds identically and carries no information. By the same lemma $\delta = U_1$, so $\delta \geq \tau \iff U \geq 0$. $\square$

9.3. Homogenization. We now express Q and U through $R(q)$. Let $q \neq 0$ and $g = |q|^{-2}R(q)$, and write R_{ij} for the entries of $R(q)$. Then $u = |q|^{-2}(R_{13}, R_{23}, R_{33})$ and $v = |q|^{-2}(R_{11}, R_{21}, R_{31})$, so that

$$m = |q|^{-2}(R_{13}, R_{23}, R_{33} - |q|^2), \quad \rho = v_0 \times v = |q|^{-2}(0, -R_{31}, R_{21}).$$

Consequently $n = |q|^{-4}n_h$, $N_2 = |q|^{-4}N_h$ and $U_1^2 = |q|^{-4}H_h$, where

$$n_h(q) = -R_{23}R_{31} + (R_{33} - |q|^2)R_{21}, \tag{32}$$

$$N_h(q) = R_{21}^2 + R_{31}^2, \tag{33}$$

$$H_h(q) = R_{23}^2 + (R_{33} - |q|^2)^2 \tag{34}$$

are homogeneous of degree 4 with integer coefficients. In the coordinates of q they simplify as follows.

Lemma 9.5. *As polynomial identities in $\mathbb{Z}[q_0, q_1, q_2, q_3]$,*

$$n_h(q) = -4q_1q_2|q|^2, \tag{35}$$

$$N_h(q) = 4(q_0^2 + q_1^2)(q_2^2 + q_3^2), \tag{36}$$

$$H_h(q) = 4\left[(q_2q_3 - q_0q_1)^2 + (q_1^2 + q_2^2)^2\right]. \tag{37}$$

Proof. From the definition of $R(q)$ in Section 9.1,

$$R_{21} = 2(q_1q_2 + q_0q_3), \quad R_{23} = 2(q_2q_3 - q_0q_1), \quad R_{31} = 2(q_1q_3 - q_0q_2), \quad R_{33} - |q|^2 = -2(q_1^2 + q_2^2).$$

Substituting these into (33) and (34) gives (36) and (37) at once, the cross terms $\pm 4q_0q_1q_2q_3$ in $R_{21}^2 + R_{31}^2$ cancelling. For (35), substituting into (32),

$$\begin{aligned} n_h &= -4(q_2q_3 - q_0q_1)(q_1q_3 - q_0q_2) - 4(q_1^2 + q_2^2)(q_1q_2 + q_0q_3) \\ &= -4(q_1q_2q_3^2 - q_0q_2^2q_3 - q_0q_1^2q_3 + q_0^2q_1q_2) - 4(q_1^3q_2 + q_0q_1^2q_3 + q_1q_2^3 + q_0q_2^2q_3) \\ &= -4q_1q_2(q_0^2 + q_1^2 + q_2^2 + q_3^2), \end{aligned}$$

the four terms carrying q_0q_3 cancelling in pairs. $\square$

The polynomial n_h^2 carries the factor $|q|^4$, which therefore divides out of the first inequality. Write $\tau^2 = A/B$ with A, B positive coprime integers and set

$$Q_h(q) = 4Bq_1^2q_2^2 - A(q_0^2 + q_1^2)(q_2^2 + q_3^2), \quad (38)$$

$$U_h(q) = 4B[(q_2q_3 - q_0q_1)^2 + (q_1^2 + q_2^2)^2] - A(|q|^2)^2. \quad (39)$$

These are homogeneous of degree 4 with integer coefficients, and both are invariant under $q \mapsto -q$. Indeed, by Lemma 9.5,

$$B(|q|^2)^4 Q = Bn_h^2 - A(|q|^2)^2 N_h = 4(|q|^2)^2 Q_h, \quad B(|q|^2)^2 U = BH_h - A(|q|^2)^2 = U_h,$$

and all the other factors are positive for $q \neq 0$. So Q and Q_h agree in sign, as do U and U_h , and Theorem 9.4 gives, for every $q \neq 0$,

$$|q|^{-2}R(q) \in \mathcal{D}(\tau) \iff Q_h(q) \geq 0 \text{ and } U_h(q) \geq 0. \quad (40)$$

We write

$$\mathcal{D}_h(\tau) = \{q \in \mathbb{R}^4 \setminus \{0\} : Q_h(q) \geq 0, U_h(q) \geq 0\},$$

a cone that is invariant under $q \mapsto \lambda q$ for every $\lambda \neq 0$. By Lemma 9.2, $\mathcal{D}_h(\tau)$ is the full preimage of $\mathcal{D}(\tau)$ under $q \mapsto |q|^{-2}R(q)$.

10. THE CERTIFICATE THEOREM

We can now state the general result. Hypotheses (H1)–(H5) below are finite assertions about rational numbers and polynomials with rational coefficients, verified by inspection, expansion, or an explicit positivity witness. Hypothesis (H6) is an inequality on a semialgebraic set. Section 13 proves it by a sum-of-squares identity.

Definition 10.1. Let $\tau \in (0, 2)$ with $\tau^2 \in \mathbb{Q}$, let $L \in \mathbb{N}$, and let $M, T^* \in \mathbb{Q}$ with $M < 0$. A *certificate at* (τ, L, M, T^*) consists of, for each $\ell = 1, \dots, L$:

- a diagonal matrix $D_\ell = \text{diag}(w_1^\ell, \dots, w_{2\ell+1}^\ell)$ with $w_i^\ell \in \mathbb{Q}_{>0}$,
- a symmetric matrix $C_\ell \in \mathbb{Q}^{(2\ell+1) \times (2\ell+1)}$, and
- a matrix T_h^ℓ of polynomials in $\mathbb{Q}[q_0, q_1, q_2, q_3]$ of size $(2\ell+1) \times (2\ell+1)$,

subject to the following hypotheses. Set

$$P(q) = \sum_{\ell=1}^L (|q|^2)^{L-\ell} \sum_{i,j=1}^{2\ell+1} \frac{(C_\ell)_{ij} (T_h^\ell)_{ij}(q)}{w_j^\ell}, \quad (41)$$

$$G(q) = M(|q|^2)^L - P(q). \quad (42)$$

(H1) (*Homogeneity.*) Every entry of T_h^ℓ is a homogeneous polynomial of degree 2ℓ .

(H2) (*Multiplicativity.*) $T_h^\ell(p * q) = T_h^\ell(p) T_h^\ell(q)$ as an identity in $\mathbb{Q}[p, q]$.

(H3) (*Orthogonality.*) $T_h^\ell(q)^\top D_\ell T_h^\ell(q) = (|q|^2)^{2\ell} D_\ell$ as an identity in $\mathbb{Q}[q]$.

(H4) (*Positive semidefiniteness.*) C_ℓ is positive semidefinite.

(H5) (*Trace.*) $\sum_{\ell=1}^L \text{tr}(D_\ell^{-1} C_\ell) = T^*$.

(H6) (*Cover.*) $G(q) \geq 0$ for every $q \in \mathcal{D}_h(\tau)$.

Theorem 10.2. If a certificate at (τ, L, M, T^*) exists, then

$$N_{\max}(\tau) \leq 1 + \left\lfloor \frac{T^*}{-M} \right\rfloor.$$

Proof. Step 1: T_h^ℓ descends to a map on $\text{SO}(3)$. For $g \in \text{SO}(3)$ choose, by Lemma 9.2(b), a unit quaternion q with $R(q) = g$, and set $T^\ell(g) = T_h^\ell(q)$. This is well defined: by Lemma 9.2(a) the only other choice is $-q$, and by (H1) the entries of T_h^ℓ are homogeneous of the even degree 2ℓ , so $T_h^\ell(-q) = T_h^\ell(q)$.

Step 2: the hypotheses of Section 7. Each D_ℓ is diagonal with positive entries, hence symmetric positive definite. Let $g_1, g_2 \in \text{SO}(3)$ and pick unit quaternions $q^{(1)}, q^{(2)}$ with $R(q^{(i)}) = g_i$. By Lemma 9.1(b), $R(q^{(1)} * q^{(2)}) = g_1 g_2$ and $|q^{(1)} * q^{(2)}| = 1$, so (H2) gives

$$T^\ell(g_1 g_2) = T_h^\ell(q^{(1)} * q^{(2)}) = T_h^\ell(q^{(1)}) T_h^\ell(q^{(2)}) = T^\ell(g_1) T^\ell(g_2),$$

and (H3) at a unit quaternion gives $T^\ell(g)^\top D_\ell T^\ell(g) = D_\ell$. Thus the data $(D_\ell, T^\ell)_{\ell=1}^L$ satisfy (i)–(ii) of Section 7, and by (H4) the sequence $\vec{C} = (C_\ell)_{\ell=1}^L$ is admissible in Definition 7.2.

Step 3: P computes the two-point sum. For a unit quaternion q we have $|q|^2 = 1$, so by (41) and Remark 7.3,

$$P(q) = \sum_{\ell=1}^L \sum_{i,j} \frac{(C_\ell)_{ij} (T_h^\ell)_{ij}(q)}{w_j^\ell} = \sum_{\ell=1}^L \text{tr} \left(D_\ell^{-1} C_\ell T^\ell(R(q)) \right).$$

By (H1) the ℓ -th summand of (41) is homogeneous of degree $2(L - \ell) + 2\ell = 2L$, so P is homogeneous of degree $2L$; hence for arbitrary $q \neq 0$,

$$\frac{P(q)}{(|q|^2)^L} = \sum_{\ell=1}^L \text{tr} \left(D_\ell^{-1} C_\ell T^\ell(|q|^{-2} R(q)) \right). \quad (43)$$

Step 4: (H6) is exactly $S(\vec{C}) \leq M$. Let $g \in \mathcal{D}(\tau)$ and pick $q \neq 0$ with $g = |q|^{-2} R(q)$. By (40) we have $q \in \mathcal{D}_h(\tau)$, so (H6) gives $G(q) \geq 0$, i.e. $P(q) \leq M(|q|^2)^L$. Dividing by $(|q|^2)^L > 0$ and using (43),

$$\sum_{\ell=1}^L \text{tr} \left(D_\ell^{-1} C_\ell T^\ell(g) \right) \leq M.$$

Taking the supremum over $g \in \mathcal{D}(\tau)$ yields $S(\vec{C}) \leq M < 0$.

Step 5: conclusion. By (H5) and Theorem 7.4,

$$N_{\max}(\tau) \leq 1 + \left\lfloor \frac{\sum_{\ell} \text{tr}(D_\ell^{-1} C_\ell)}{-M} \right\rfloor = 1 + \left\lfloor \frac{T^*}{-M} \right\rfloor. \quad \square$$

In the language of Definition 1.4, the conclusion of Theorem 10.2 reads $(1 + \lfloor T^*/(-M) \rfloor, \tau/(2 - \tau)) \in \mathcal{P}_{\text{cyl}}$.

11. PROOF OF THEOREM 1.6

The five certificates of Table 2 prove Theorem 1.6 and Corollaries 1.9 and 1.10. They share the matrices D_ℓ and T_h^ℓ of Section 12 and differ in τ^2 , L , the matrices C_ℓ , the constants M and T^* , and the sum-of-squares data of Section 13.

certificate	τ	$\omega = \frac{\tau}{2-\tau}$	L	proves
(A) cyl_N3_omega91_37	$\frac{91}{64}$	$\frac{91}{37}$	6	$(3, \frac{91}{37}) \in \mathcal{P}_{\text{cyl}}$
(B) cyl_N4_omega39_25	$\frac{39}{32}$	$\frac{39}{25}$	4	$(4, \frac{39}{25}) \in \mathcal{P}_{\text{cyl}}$
(C) cyl_N5_omega1203_845	$\frac{1203}{1024}$	$\frac{1203}{845}$	4	$(5, \frac{1203}{845}) \in \mathcal{P}_{\text{cyl}}$
(D) cyl_N6_omega1_1	1	1	4	$(6, 1) \in \mathcal{P}_{\text{cyl}}$
(E) cyl_N7_omega14558721_18995711	$\frac{14558721}{2^{24}}$	$\frac{14558721}{18995711}$	6	$(7, \frac{14558721}{18995711}) \in \mathcal{P}_{\text{cyl}}$

TABLE 2. The deposited certificates. Each file is named after the statement it proves.

Proof of Corollary 1.10. Certificate (A) is a certificate at $\tau = \frac{91}{64}$ (Table 2). Its hypotheses are verified in Sections 12 and 13, and $T^* < 3(-M)$ is an exact rational comparison performed by the verifier. Theorem 10.2 gives $N_{\max}(\frac{91}{64}) \leq 3$, i.e. $N(\omega) \leq 3$ for every $\omega \geq \frac{91}{37}$ by (3) and Lemma 2.5. Hence $R_4 \leq \frac{91}{37} = 2.459\dots$, and with the ring of four, $1 + \sqrt{2} = 2.414\dots \leq R_4$, the first decimal of R_4 is 4. $\square$

Proof of Corollary 1.9. Certificate (A) gives $N_{\max}(\frac{91}{64}) \leq 3$ (Corollary 1.10). Now let $\omega \geq 3$ and put $\tau = \frac{2\omega}{1+\omega}$. The map $\omega \mapsto \frac{2\omega}{1+\omega}$ is increasing, so $\tau \geq \frac{3}{2} > \frac{91}{64}$, and Lemma 2.5 gives

$$N(\omega) = N_{\max}(\tau) \leq N_{\max}(\frac{91}{64}) \leq 3.$$

Thus at most three pairwise non-overlapping cylinders of radius ω touch the unit ball, for every $\omega \geq 3$. In particular $N(3) \leq 3 < 4$, so by Corollary 2.7 $R_4 < 3$. $\square$

Proof of Theorem 1.6. The lower bounds $N(\omega_n) \geq n$ are the ring. For the upper bounds:

$n = 3$. Certificate (A) at $\tau = \frac{91}{64}$ has $1 + \lfloor T^*/(-M) \rfloor = 3$, so $(3, \frac{91}{37}) \in \mathcal{P}_{\text{cyl}}$; and $\frac{91}{37} < 3 < 3 + 2\sqrt{3} = \omega_3$. Alternatively, the elementary bound $R_4 < 2 + \sqrt{6}$ of Remark 11.2 already suffices for this case, since $2 + \sqrt{6} < 3 + 2\sqrt{3} = \omega_3$.

$n = 4$. Certificate (B), at $\tau = \frac{39}{32}$, gives $N_{\max}(\frac{39}{32}) \leq 4$ by Theorem 10.2, i.e. $(4, \frac{39}{25}) \in \mathcal{P}_{\text{cyl}}$, and $\frac{39}{25} \leq 1 + \sqrt{2} = \omega_4$ since $14^2 = 196 \leq 1250 = 2 \cdot 25^2$.

$n = 5$. Certificate (C), at $\tau = \frac{1203}{1024}$, gives $N_{\max}(\frac{1203}{1024}) \leq 5$, i.e. $(5, \frac{1203}{845}) \in \mathcal{P}_{\text{cyl}}$. It remains to check $\omega_5 \geq \frac{1203}{845}$, i.e. $\sin \frac{\pi}{5} \geq \frac{1203}{2048}$. Squaring and using $16 \sin^2 \frac{\pi}{5} = 10 - 2\sqrt{5}$, this is equivalent in turn to

$$10 - 2\sqrt{5} \geq \frac{1447209}{262144} \iff 1174231 \geq 524288\sqrt{5} \iff 1174231^2 \geq 5 \cdot 524288^2,$$

and indeed $1378818441361 \geq 1374389534720$.

$n = 6$. Certificate (D) (and, independently, [13]) gives $N_{\max}(1) \leq 6$; since $2 \sin \frac{\pi}{6} = 1$ corresponds to $\omega = \omega_6 = 1$ exactly, $(6, \omega_6) \in \mathcal{P}_{\text{cyl}}$.

$n = 7$. Certificate (E), at $\tau = \frac{14558721}{2^{24}}$, gives $N_{\max}(\tau) \leq 7$, i.e. $(7, \frac{\tau}{2-\tau}) \in \mathcal{P}_{\text{cyl}}$. It remains to check $\omega_7 \geq \frac{\tau}{2-\tau}$, i.e. $\sin \frac{\pi}{7} \geq \frac{\tau}{2}$. Since $\sin^2 \frac{\pi}{7} = \frac{1}{2}(1 - \cos \frac{2\pi}{7})$ and $c = \cos \frac{2\pi}{7}$ is the largest root of $p(x) = 8x^3 + 4x^2 - 4x - 1$, the inequality is equivalent to $c \leq r := 1 - \frac{\tau^2}{2}$. Since p is increasing to the right of its largest root and $r > 0.62$ exceeds the other two roots, this is equivalent to $p(r) \geq 0$. With $\tau^2 = \frac{211956357155841}{2^{48}}$, $p(r) > 0$ is a comparison of two integers, which holds. $\square$

Remark 11.1 (Attribution for $n = 4$). The membership $(4, \omega_4) \in \mathcal{P}_{\text{cyl}}$ is equivalent to $R_5 < 1 + \sqrt{2}$, and this was first established by Firsching [5], building on Braß–Wenk [3], in the stronger form $R_5 \leq 1.89395$; see also the account in [14, Section 6]. Certificate (B) provides an independent proof by a different method, with the sharper exact bound $R_5 \leq \frac{39}{25} = 1.56$. For comparison, the ring gives $R_5 \geq \omega_5 = 1.4259 \dots$

Remark 11.2. The elementary bound of Section 1.4 is obtained as follows. The tangency points u_i of the rescaled axes satisfy $\text{dist}(\ell_i, \ell_j) \leq |u_i - u_j|$, so tangent lines at pairwise distances $\geq \tau$ give points on the unit sphere at pairwise angles $\geq 2 \arcsin \frac{\tau}{2}$; four unit vectors always have some pairwise angle at most $\arccos(-\frac{1}{3})$, which yields $\tau \leq \frac{2\sqrt{6}}{3} = 1.6329 \dots$, i.e. $R_4 < 2 + \sqrt{6}$, and five unit vectors have some pairwise angle at most $\frac{\pi}{2}$, which yields $\tau \leq \sqrt{2}$, i.e. $R_5 < 1 + \sqrt{2}$. The gaps between these values and the certified separations of Table 2 measure what the two-point bound sees and the projection argument does not, namely the constraint imposed by the infinite length of the cylinders.

12. THE CERTIFICATE DATA

This section describes the deposited data and the checks that establish (H1)–(H5). How the data was found is described in Section 14 and plays no role here. The *universal* part, shared by all certificates, consists, for each $\ell = 1, \dots, 6$, of:

- positive rationals $w_1^\ell, \dots, w_{2\ell+1}^\ell$, forming the diagonal matrix $D_\ell = \text{diag}(w_1^\ell, \dots, w_{2\ell+1}^\ell)$;
- an integer table defining a $(2\ell + 1) \times (2\ell + 1)$ matrix $T_h^\ell(q)$ of polynomials in $q = (q_0, q_1, q_2, q_3)$, with coefficients in $\frac{1}{\text{Tden}_\ell} \mathbb{Z}$ for a deposited scale $\text{Tden}_\ell \in \{1, 2, 4\}$ (the deposited values are 1, 1, 2, 1, 4, 2 for $\ell = 1, \dots, 6$).

A certificate with truncation L uses the first L universal blocks; the deposited certificates have $L = 4$ or $L = 6$. Each certificate additionally deposits the rational τ^2 , the constants M and T^* , and symmetric rational matrices $C_\ell \in \mathbb{Q}^{(2\ell+1) \times (2\ell+1)}$ together with integer witnesses for their positive semidefiniteness.

Positivity of each w_i^ℓ , required by Definition 10.1, is read off from the deposited numerators and denominators. Hypotheses (H1), (H2), and (H3) are finite computations on the universal tables, performed once for all certificates. Hypothesis (H4) follows from the deposited witnesses by Proposition 12.2, and (H5) is a finite computation per certificate. Hypothesis (H6) is the subject of Section 13.

12.1. **The identities (H1)–(H3).** Hypothesis (H1) is an inspection of the exponents in the deposited table: every row of T_h^ℓ carries exponents summing to 2ℓ .

Both (H2) and (H3) are identities between polynomials with integer coefficients once the deposited scales are cleared. Writing $A^\ell = \text{Tden}_\ell T_h^\ell$, which has integer entries, and $\widehat{D}_\ell = m_\ell D_\ell$, where m_ℓ is a common denominator of $w_1^\ell, \dots, w_{2\ell+1}^\ell$, they read

$$\text{Tden}_\ell A^\ell(p * q) = A^\ell(p) A^\ell(q), \quad A^\ell(q)^\top \widehat{D}_\ell A^\ell(q) = \text{Tden}_\ell^2 |q|^{4\ell} \widehat{D}_\ell,$$

so each amounts to comparing two polynomial expansions with integer coefficients, coefficient by coefficient.

12.2. **Positive semidefiniteness (H4).** The positive semidefiniteness of the matrices $C_1, \dots, C_L$ of each certificate follows from the deposited integer witnesses by the following two elementary facts.

Lemma 12.1. *Let $E \in \mathbb{R}^{n \times n}$ be symmetric with $E_{ii} \geq \sum_{j \neq i} |E_{ij}|$ for every i . Then $E \succeq 0$.*

Proof. Let $v \in \mathbb{R}^n$. Using $|v_i v_j| \leq \frac{1}{2}(v_i^2 + v_j^2)$ and then the symmetry of E ,

$$\left| \sum_{i \neq j} E_{ij} v_i v_j \right| \leq \sum_{i \neq j} |E_{ij}| \frac{1}{2}(v_i^2 + v_j^2) = \sum_i v_i^2 \sum_{j \neq i} |E_{ij}| \leq \sum_i E_{ii} v_i^2,$$

so $v^\top E v = \sum_i E_{ii} v_i^2 + \sum_{i \neq j} E_{ij} v_i v_j \geq 0$. $\square$

Proposition 12.2. *Write $\widehat{C}_\ell := 2^\kappa C_\ell$ for the deposited integer matrix, where κ is the even integer deposited as cDenLog2 , and let S_ℓ and W_ℓ be the deposited positive integer scalar and integer matrix. If*

$$E_\ell = S_\ell \widehat{C}_\ell - W_\ell^\top W_\ell$$

is symmetric and diagonally dominant, in the sense that $(E_\ell)_{ii} \geq \sum_{j \neq i} |(E_\ell)_{ij}|$ for every i , then C_ℓ is symmetric positive semidefinite.

Proof. By Lemma 12.1, $E_\ell \succeq 0$. Also $W_\ell^\top W_\ell \succeq 0$, since $v^\top W_\ell^\top W_\ell v = |W_\ell v|^2 \geq 0$ for every v . Hence

$$2^\kappa S_\ell C_\ell = S_\ell \widehat{C}_\ell = E_\ell + W_\ell^\top W_\ell \succeq 0,$$

and dividing by the positive scalar $2^\kappa S_\ell$ gives $C_\ell \succeq 0$. Symmetry of C_ℓ is symmetry of the deposited $\widehat{C}_\ell$. $\square$

12.3. **The trace sum (H5).** The verifier evaluates $\sum_\ell \sum_i (C_\ell)_{ii} / w_i^\ell$ exactly and checks that it equals the deposited T^* . This is (H5).

12.4. **The certificate polynomial.** The polynomial P of Definition 10.1, formed from the deposited C_ℓ , the tables T_h^ℓ and the weights, has rational coefficients and is homogeneous of degree $2L$: by (H1) each $T_h^\ell(q)_{ij}$ is homogeneous of degree 2ℓ , and $2(L - \ell) + 2\ell = 2L$ for every ℓ . The same holds for $G = M(|q|^2)^L - P$.

13. A SUM-OF-SQUARES PROOF OF HYPOTHESIS (H6)

By Theorem 10.2 and Section 12, the proof of each corollary is complete once (H6) is established for the corresponding certificate, that is, once we show

$$G(q) \geq 0 \quad \text{on} \quad \mathcal{D}_h(\tau) = \{q \neq 0 : Q_h(q) \geq 0 \text{ and } U_h(q) \geq 0\}, \quad (44)$$

with Q_h, U_h formed from the certificate's $\tau^2 = A/B$ by (38)–(39).

The statement asks that one polynomial be non-negative wherever two others are. The certificate proves it by an identity that writes a positive multiple of G as $\sigma_0 + \sigma_1 U_h + \sigma_2 Q_h + \sigma_3 U_h Q_h$, where each σ_a is a sum of squares of polynomials. Such a σ_a is non-negative everywhere, so the right-hand side is non-negative wherever U_h and Q_h are. Given such an identity, (44) follows in two lines (Theorem 13.2), and verifying the identity is a computation of the kind the verifier already performs for (H2) and (H3): expand both sides and compare integer coefficients. The method of certifying polynomial inequalities on semialgebraic sets by sums of squares goes back to the Positivstellensätze of real algebraic geometry and to its computational development by Lasserre and Parrilo [10, 19].

13.1. Sums of squares, and how a matrix certifies one. Fix an integer $d \geq 0$ and let $\mathbf{m}_d(q)$ denote the column vector of all $n_d = \binom{d+3}{3}$ monomials $q_0^{e_0} q_1^{e_1} q_2^{e_2} q_3^{e_3}$ of total degree d , listed in a fixed order. The verifier uses the lexicographic order of the exponent vectors.

Lemma 13.1. *Let $\Sigma \in \mathbb{R}^{n_d \times n_d}$ be symmetric positive semidefinite, and define*

$$\sigma(q) = \mathbf{m}_d(q)^\top \Sigma \mathbf{m}_d(q).$$

Then σ is a homogeneous polynomial of degree $2d$ and $\sigma(q) \geq 0$ for every $q \in \mathbb{R}^4$. Moreover σ is a sum of squares of forms of degree d .

Proof. Each entry of $\mathbf{m}_d(q)\mathbf{m}_d(q)^\top$ is a monomial of degree $2d$, so σ is a form of degree $2d$. For fixed q the number $\sigma(q) = v^\top \Sigma v$ with $v = \mathbf{m}_d(q)$ is non-negative because $\Sigma \succeq 0$. For the last statement, write $\Sigma = W^\top W$ (for instance from the spectral decomposition); then $\sigma(q) = |W\mathbf{m}_d(q)|^2$ is the sum of the squares of the forms $(W\mathbf{m}_d(q))_i$. $\square$

Only the inequality $\sigma \geq 0$ is used below, and it needs only $\Sigma \succeq 0$. Positive semidefiniteness of a rational Σ is established from a deposited integer witness by Proposition 12.2, exactly as for the matrices C_ℓ .

13.2. The certificate identity.

Theorem 13.2. *Fix a certificate at (τ, L, M, T^*) , let G be its cover polynomial (42), and let Q_h, U_h be as in (38)–(39). Suppose we are given an integer $k \geq 0$ with $L + k \geq 4$ and four symmetric positive semidefinite matrices with rational entries,*

$$\Sigma_0 \in \mathbb{Q}^{n_{L+k} \times n_{L+k}}, \quad \Sigma_1, \Sigma_2 \in \mathbb{Q}^{n_{L+k-2} \times n_{L+k-2}}, \quad \Sigma_3 \in \mathbb{Q}^{n_{L+k-4} \times n_{L+k-4}},$$

such that, with $\sigma_a = \mathbf{m}^\top \Sigma_a \mathbf{m}$ formed from the monomial vector of the matching degree, the identity

$$(|q|^2)^k G = \sigma_0 + \sigma_1 U_h + \sigma_2 Q_h + \sigma_3 U_h Q_h \quad (45)$$

holds in $\mathbb{Q}[q_0, q_1, q_2, q_3]$. Then hypothesis (H6) holds.

Proof. Let $q \in \mathcal{D}_h(\tau)$, so that $U_h(q) \geq 0$ and $Q_h(q) \geq 0$. By Lemma 13.1 each $\sigma_a(q) \geq 0$, so every one of the four summands on the right-hand side of (45) is non-negative at q , and therefore $(|q|^2)^k G(q) \geq 0$. Since $q \neq 0$, dividing by $(|q|^2)^k > 0$ gives $G(q) \geq 0$. $\square$

Remark 13.3. Both sides of (45) are forms of degree $2L + 2k$: the left by Section 12.4, and on the right σ_0 has degree $2(L + k)$, while $\sigma_1 U_h, \sigma_2 Q_h$ and $\sigma_3 U_h Q_h$ have degrees $2(L + k - 2) + 4$ and $2(L + k - 4) + 4 + 4$, the generators being of degree 4 by Lemma 9.5. The comparison therefore takes place in the single space of forms of degree $2L + 2k$, of dimension n_{2L+2k} . Any of $\Sigma_0, \dots, \Sigma_3$ may be the zero matrix.

Remark 13.4 (A worked miniature). The following two-variable example contains the entire mechanism. Let

$$F(x, y) = 2x^4 - 3x^2y^2 + y^4, \quad u(x, y) = x^2 - y^2.$$

The form F is not non-negative on $\mathbb{R}^2$: at $(x, y) = (\sqrt{2}, \sqrt{3})$ it takes the value $8 - 18 + 9 = -1$. Nevertheless $F \geq 0$ wherever $u \geq 0$, and the one-line proof is the identity

$$F = (x^2 - y^2)^2 + x^2 u,$$

which the reader checks by expanding. On $\{u \geq 0\}$ the right-hand side is a sum of two non-negative terms, and hence it is non-negative.

13.3. What is deposited, and how it is verified. For each certificate the following records are deposited (the format is that of Appendix B): the integer k of (45) as `sosExponentK`; an even integer κ_s as `sosDenLog2`; the integer symmetric matrices `sosGram000a` $= 2^{\kappa_s} \Sigma_a$ for $a = 0, \dots, 3$; and, for each a , the integer witnesses `sosWit000a` and `sosWit000a` of Proposition 12.2. The verifier then performs the following steps, all in exact arithmetic.

- (S1) *Shape.* Each deposited $2^{\kappa_s} \Sigma_a$ is symmetric and of the size prescribed by Theorem 13.2, namely $n_{L+k}, n_{L+k-2}, n_{L+k-2}$ and n_{L+k-4} , computed from the deposited L and k .
- (S2) *Positivity.* For each a , the matrix $E_a = S_a(2^{\kappa_s} \Sigma_a) - W_a^\top W_a$ is symmetric and diagonally dominant, whence $\Sigma_a \succeq 0$ by Proposition 12.2.
- (S3) *Assembly.* The polynomial G is built from the deposited C_ℓ , the universal tables and the weights, as in Section 12.4. The generators Q_h and U_h are read from the certificate records and validated against (38)–(39) at the deposited A and B (check (V2) of Appendix B). The left-hand side of (45) is G multiplied k times by $|q|^2$.

(S4) *The identity.* The right-hand side of (45) is expanded: for each a and each pair $i \leq j$ of indices, the entry $(\Sigma_a)_{ij}$ contributes $(2 - \delta_{ij})(\Sigma_a)_{ij}$ times the monomial $(\mathbf{m})_i(\mathbf{m})_j$, and the blocks $a = 1, 2, 3$ are then multiplied by U_h , Q_h and $U_h Q_h$ respectively. Both sides are scaled by a common denominator 2^{K_s} . The identity becomes an equality between two lists of n_{2L+2k} integers, compared entry by entry.

If (S1)–(S4) succeed, then Theorem 13.2 applies and (H6) holds. The number of coefficient equations in (S4) is $n_{2L+2k} = \binom{2L+2k+3}{3}$. All intermediate integers are bounded by an explicit function of the deposited scales and are checked against the machine word width as in Appendix B. A reader who wishes to reimplement the check needs only a parser for the format of Appendix B and exact polynomial multiplication in four variables.

Remark 13.5 (Existence). Identities of the shape (45) exist whenever $G > 0$ on the compact set $\mathcal{D}_h(\tau) \cap \mathbb{S}^3$: Schmüdgen’s Positivstellensatz [21] then gives a representation of G in the preordering generated by U_h , Q_h and $|q|^2 = 1$, which homogeneity brings to the form (45) for some k . The theorem bounds neither k nor the degree, and the deposited identities were found by the semidefinite programs of Section 14.4.

Hypotheses (H1)–(H6) are established for each of the deposited certificates, so Theorem 10.2 yields the bounds of Table 2, and with them Theorem 1.6 and the corollaries of Section 1.

14. HOW THE CONFIGURATIONS AND CERTIFICATES WERE FOUND

This section describes how we found the configurations of Section 6 and the certificates of Section 12.

14.1. The window configurations. The configurations for $n \in \{9, 10, 11, 12\}$ were found among configurations with a prescribed symmetry group. The method described next also finds configurations for $n \geq 10$, but the symmetric ones have larger minimum distances and simpler pictures. For $n \geq 13$ the configurations come from numerical optimization. Each of the $N = n + 1$ lines is parametrized by its tangency point and a tangent direction, and we maximize the minimum pairwise distance in three stages, each starting from the output of the previous one. First, for an increasing sequence of targets t up to $1.02 \cdot 2 \sin \frac{\pi}{n}$, we minimize the penalty $\sum_{i < j} \max(0, t - d_{ij})^2$, which vanishes exactly when all pairs are at distance at least t . Second, we maximize the smooth lower bound $-\beta^{-1} \log \sum_{i < j} e^{-\beta d_{ij}}$ for $\beta = 200, 1000, 5000$. Third, we maximize t subject to $d_{ij} \geq t$ for all $i < j$ by sequential quadratic programming.

Random initial configurations do not work. At $N = 20$ such runs end with a minimum distance below that of the ring. The reason is that the distance between two tangent lines is discontinuous at parallel pairs. If one of two parallel lines at distance d is tilted by an arbitrarily small angle, the distance jumps to $d |\sin \varphi|$, where φ is the angle between the tilt direction and the common perpendicular. Started from a generic configuration, the optimization does not find the nearly parallel neighbours that the ring and the pinwheel have. We therefore started from configurations that already have such neighbours, such as pinwheels and configurations found for nearby values of N with a line added or removed, and improved the best configuration found for a given N by random perturbation followed by the three stages.

The ten-line configuration for $n = 9$ is a rounded version of a symmetric configuration. Its symmetry group is the Klein four-group $\mathbb{D}_2$, realized as the rotations by π about three mutually perpendicular axes. The ten lines form three orbits. Two orbits (green and red in the first picture of Figure 4) are obtained by starting with two lines in general position. This gives eight lines. The remaining two lines (blue) form one orbit of a special kind: start with a line tangent to $\mathbb{S}^2$ at $(1, 0, 0)$. The rotation by π about that axis maps this line to itself, whatever its direction, and the other two rotations map it to a tangent line at $(-1, 0, 0)$. So its orbit has two elements. Up to a global rotation the configuration depends on seven parameters, three for each general orbit and one for the direction of the axial pair, instead of the $3N - 3 = 27$ of an arbitrary configuration of ten lines. This configuration is simpler than those found by optimization, yet it was the last one we found. A search over all 27 parameters does not find it, while an optimization restricted to the seven parameters finds it quickly.

The twelve-line configuration for $n = 11$ is a rational approximation of the following one. Take the tangent line at $\frac{1}{\sqrt{10}}(3, 1, 0)$ with direction $(0, 0, 1)$ and its orbit under the rotation group $\mathbb{A}_4$ of a regular tetrahedron, generated by the rotations by π about the coordinate axes and the cyclic permutation of the coordinates. The twelve lines are parallel to the coordinate axes, four to each, and their minimal distance is $2/\sqrt{10} > 2 \sin \frac{\pi}{10}$. Hence twelve cylinders of radius $1/(\sqrt{10}-1) > \omega_{10}$ touch the ball, and this configuration alone proves $(10, \omega_{10}) \notin \mathcal{P}_{\text{cyl}}$ and $(11, \omega_{11}) \notin \mathcal{P}_{\text{cyl}}$. Ogievetsky and Shlosman [17] study configurations with Platonic symmetry, obtained by rotating the edges of a Platonic solid about the radii through their midpoints. For the tetrahedron this gives Kuperberg’s O_6 , and for the octahedron twelve cylinders of radius $7 - \sqrt{2} - 4\sqrt{3} + 3\sqrt{6} = 0.349\dots$, which is exactly ω_{12} . The configuration here has a smaller symmetry group, a generic orbit as its set of tangency points, and a larger radius.

14.2. The representations and the tables. The irreducible real representations of $\text{SO}(3)$ are its actions on the spaces H_ℓ of harmonic polynomials on $\mathbb{R}^3$, homogeneous of degree ℓ , of dimension $2\ell + 1$. Bochner's theorem for compact groups identifies the continuous functions of positive type on $\text{SO}(3)$ as the sums

$$g \mapsto \sum_{\ell \geq 0} \text{tr}(A_\ell \pi^\ell(g)), \quad A_\ell \succeq 0,$$

over those representations π^ℓ . Truncating at $\ell \leq L$ turns the search for a bound into a finite semidefinite program in the matrices A_ℓ . The tables T_h^ℓ are these representations in an explicit basis, and the matrices C_ℓ are the A_ℓ .

The basis is obtained as follows. Fix ℓ , take any basis of H_ℓ with integer coefficients, and apply Gram–Schmidt with respect to $\langle p, r \rangle = \frac{1}{4\pi} \int_{\mathbb{S}^2} p r d\sigma$. Every such inner product is rational, since

$$\frac{1}{4\pi} \int_{\mathbb{S}^2} x^a y^b z^c d\sigma = \begin{cases} \frac{(a-1)!!(b-1)!!(c-1)!!}{(a+b+c+1)!!}, & a, b, c \text{ all even,} \\ 0, & \text{otherwise,} \end{cases}$$

so the process never leaves $\mathbb{Q}$. Clearing denominators gives integer polynomials $h_1^\ell, \dots, h_{2\ell+1}^\ell$ with $\langle h_i^\ell, h_j^\ell \rangle = \delta_{ij} w_i^\ell$, $w_i^\ell > 0$. This is the matrix D_ℓ . For $g = R(q)$ the coefficients in

$$h_j^\ell(R(q)^\top x) = \sum_{i=1}^{2\ell+1} T_h^\ell(q)_{ij} h_i^\ell(x) \quad (46)$$

are polynomials in q , and these are the deposited tables. From (46), (H1) is a degree count, since the entries of $R(q)$ are quadratic in q , (H2) is the composition law of the action, and (H3) is the invariance of σ under $\text{SO}(3)$. The harmonic polynomials themselves are not deposited.

14.3. The matrices C_ℓ and the constants. The C_ℓ are the free variables of the semidefinite program, solved once per value of τ . A numerical solution is rounded to rational values, and positive semidefiniteness, the trace sum and the bound on $\mathcal{D}$ are then established for the rounded matrices by Sections 12 and 13.

Larger L gives a bound at least as good. $L = 4$ suffices at $\tau = 1$, while the certificates closest to their thresholds, for $n = 3$ and $n = 7$, use $L = 6$. For M there is an interval of admissible values, $S(\vec{C}) \leq M < -T^*/\text{targetBound}$, the left end being what Section 13 certifies and the right end the counting inequality. The deposited M lie strictly inside this interval, because the rounding of Section 14.4 needs the identity (45) to have an interior solution.

14.4. The multipliers Σ_a . The matrices Σ_a of Theorem 13.2 are found in two stages. For a fixed k , the identity (45) is a system of n_{2L+2k} linear equations in the entries of the Σ_a , one per monomial of degree $2L + 2k$, and the constraints $\Sigma_a \succeq 0$ make it a semidefinite feasibility problem [10, 19]. It is solved numerically with an objective that pushes the solution into the interior of the cone, for instance by maximizing the least eigenvalue of the Σ_a subject to a norm bound.

The second stage makes the solution exact, following Peyrl and Parrilo [20]. The floating-point Σ_a are rounded to dyadic rationals with denominator 2^{k_s} and then projected, in exact rational arithmetic, onto the affine subspace of tuples satisfying (45). This is a rational least-squares step, since the subspace is cut out by linear equations with rational coefficients. If the numerical solution was interior by more than the combined rounding and projection error, the projected matrices are positive definite, and integer witnesses in the sense of Proposition 12.2 are produced for them.

15. THE SECOND DECIMAL AND THE THREE-POINT BOUND

Corollary 1.10 determines the first decimal digit of R_4 . By (3), pinning $R_4 = 2.41 \dots$ would require a certificate with $\text{targetBound}=3$ at a separation τ with $\sqrt{2} < \tau < \frac{242}{171}$, and the full Conjecture 1.3 would require one at $\tau = \sqrt{2}$ itself. We did not find either. This section states what such a certificate would have to look like. At two points (Theorem 15.1) the answer is rigid in a specific way. At three points (Theorem 15.8) the corresponding condition is strictly weaker. Nothing here is used in the proofs of the results stated in the introduction.

15.1. A sharp two-point certificate.

Theorem 15.1. *Let $L \in \mathbb{N}$, let $M < 0$, and suppose that for $\ell = 1, \dots, L$ we are given symmetric positive-definite matrices $D_\ell \in \mathbb{R}^{n_\ell \times n_\ell}$, symmetric positive-semidefinite matrices $C_\ell \in \mathbb{R}^{n_\ell \times n_\ell}$, and maps $T^\ell : \text{SO}(3) \rightarrow \mathbb{R}^{n_\ell \times n_\ell}$ satisfying, for all $g, g_1, g_2 \in \text{SO}(3)$,*

$$T^\ell(g_1 g_2) = T^\ell(g_1) T^\ell(g_2) \quad \text{and} \quad T^\ell(g)^\top D_\ell T^\ell(g) = D_\ell.$$

Write $F(g) = \sum_{\ell=1}^L \text{tr}(D_\ell^{-1} C_\ell T^\ell(g))$ and assume:

- (i) $F(g) \leq M$ for every $g \in \mathcal{D}(\sqrt{2})$;
- (ii) $\sum_{\ell=1}^L \text{tr}(D_\ell^{-1} C_\ell) = -3M$;
- (iii) the level set $Z = \{g \in \mathcal{D}(\sqrt{2}) : F(g) = M\}$ satisfies

$$Z \subseteq \{g : \text{dist}(\ell_0, g \cdot \ell_0) = \sqrt{2}\} \cup \{g : \text{dist}(\ell_0, g \cdot \ell_0) = 2\}.$$

Then $R_4 \leq 1 + \sqrt{2}$, and consequently, by (2), $R_4 = 1 + \sqrt{2}$.

Conditions (i) and (ii) say that the trace sum bounds $N_{\max}(\sqrt{2})$ by exactly 3. Condition (iii) says that the bound is attained only on the two distance strata of the ring of four. The hypotheses are stated over the reals, with no rationality or truncation restrictions.

Lemma 15.2. *Under the hypotheses of Theorem 15.1, for all $g_1, \dots, g_N \in \text{SO}(3)$ one has $\sum_{i,j=1}^N F(g_i^{-1} g_j) \geq 0$, and $F(e) = \sum_\ell \text{tr}(D_\ell^{-1} C_\ell)$.*

Proof. As in the proof of Theorem 7.4: by Lemma 7.1, $T^\ell(g_i^{-1} g_j) = D_\ell^{-1} T^\ell(g_i)^\top D_\ell T^\ell(g_j)$, so with $S_\ell = \sum_i T^\ell(g_i)$ the double sum equals $\sum_\ell \text{tr}((D_\ell^{-1} C_\ell D_\ell^{-1})(S_\ell^\top D_\ell S_\ell))$, a sum of traces of products of pairs of positive-semidefinite matrices. The second assertion is $T^\ell(e) = I$ (Lemma 7.1). $\square$

Lemma 15.3. *If ℓ_1, ℓ_2 are lines tangent to the unit sphere at u_1, u_2 , then $\text{dist}(\ell_1, \ell_2) \leq 2$, and $\text{dist}(\ell_1, \ell_2) = 2$ forces $u_2 = -u_1$. Consequently no three tangent lines have all pairwise distances equal to 2.*

Proof. Since $u_i \in \ell_i$, we have $\text{dist}(\ell_1, \ell_2) \leq |u_1 - u_2| \leq |u_1| + |u_2| = 2$. If $\text{dist}(\ell_1, \ell_2) = 2$, both inequalities are equalities, and equality in the triangle inequality for unit vectors gives $u_2 = -u_1$. For the last assertion, pairwise antipodality of u_1, u_2, u_3 would give $u_3 = -u_1$ and $u_3 = -u_2 = u_1$, forcing $u_1 = 0$. $\square$

Proof of Theorem 15.1. Suppose, for contradiction, that four pairwise non-overlapping cylinders of some radius $\omega > 1 + \sqrt{2}$ touch the unit ball. By Section 2 there are distinct $g_1, \dots, g_4 \in \text{SO}(3)$ with

$$\text{dist}(\ell_0, g_i^{-1} g_j \cdot \ell_0) \geq \tau(\omega) = \frac{2\omega}{1+\omega} > \sqrt{2} \quad (i \neq j), \quad (47)$$

so in particular $g_i^{-1} g_j \in \mathcal{D}(\sqrt{2})$ for all $i \neq j$.

By Lemma 15.2 and hypothesis (ii),

$$0 \leq \sum_{i,j=1}^4 F(g_i^{-1} g_j) = 4 \cdot (-3M) + \sum_{i \neq j} F(g_i^{-1} g_j) \leq -12M + 12M = 0,$$

where the last inequality applies (i) to each of the twelve off-diagonal terms. Hence every inequality above is an equality. In particular the twelve numbers $F(g_i^{-1} g_j) \leq M$ sum to $12M$, so

$$F(g_i^{-1} g_j) = M \quad \text{for all } i \neq j,$$

that is, $g_i^{-1} g_j \in Z$ for all $i \neq j$. By hypothesis (iii), each distance $\text{dist}(\ell_0, g_i^{-1} g_j \cdot \ell_0)$ equals $\sqrt{2}$ or 2. By (47) it exceeds $\sqrt{2}$, so all twelve distances equal 2. The rescaled axes are then four tangent lines with all pairwise distances 2, contradicting Lemma 15.3.

Therefore $\mathcal{N}(\omega) \leq 3$ for every $\omega > 1 + \sqrt{2}$, so $R_4 = \sup\{\omega : \mathcal{N}(\omega) \geq 4\} \leq 1 + \sqrt{2}$. The ring of four (2) gives the reverse inequality. $\square$

Remark 15.4. Hypothesis (iii) cannot be strengthened by deleting the stratum $\{\text{dist} = 2\}$. The ring of four exists at separation $\sqrt{2}$ with distance spectrum $\{\sqrt{2}, 2\}$, adjacent and opposite pairs, and feeding it into Lemma 15.2 under (i) and (ii) gives $0 \leq -12M + 8F(\text{adj}) + 4F(\text{opp})$ with every value $\leq M$. This forces $F = M$ on all its relative rotations, including the opposite pairs at distance 2. The same computation shows that (iii) contains a rigidity assertion: every four-line configuration with minimal separation $\sqrt{2}$ must have distance spectrum inside $\{\sqrt{2}, 2\}$, since each of its relative rotations is forced into Z .

The rigidity of Remark 15.4 is the obstacle. A two-point trace sum that is sharp at $\tau = \sqrt{2}$ must be extremal on a set that contains all relative rotations of the ring of four and nothing outside two distance strata. We expect that no finite trace sum has this property. We also expect that the two-point bound does not decide the second decimal of R_4 at any truncation. The natural next tool is the three-point bound.

15.2. The three-point bound. The bound of Section 7 uses, of a non-overlapping set X , only the fact that each pair of its elements lies in $\mathcal{D}$. The remedy, introduced for binary codes by Schrijver [22] and for spheres by Bachoc and Vallentin [1], is to use kernels on triples; the general hierarchy on compact spaces is due to de Laat and Vallentin [8]. See [4, 9] for three-point bounds in other settings. We state the three-point bound directly on the group, in the same form as Theorem 7.4.

Throughout, $\tau \in (0, 2)$ is fixed and $\mathcal{D} = \mathcal{D}(\tau)$. The group $\text{SO}(3)$ acts on $\text{SO}(3)^3$ diagonally, $h \cdot (x_1, x_2, x_3) = (hx_1, hx_2, hx_3)$. For $x \in \text{SO}(3)^3$ let $\pi(x)$ be the set partition of $\{1, 2, 3\}$ into the classes of the relation $x_i = x_j$. There are five such partitions: $\{123\}$, the three partitions with one block of size two, and $\{1\}\{2\}\{3\}$. For a partition π write $|\pi|$ for its number of blocks and $(N)_k = N(N-1) \cdots (N-k+1)$.

Definition 15.5 (Strata). For a set partition π of $\{1, 2, 3\}$ let

$$\mathcal{D}_\pi = \{x \in \text{SO}(3)^3 : \pi(x) = \pi \text{ and } x_i^{-1}x_j \in \mathcal{D} \text{ whenever } i, j \text{ lie in different blocks of } \pi\}.$$

A *three-point positive kernel* is a continuous function $K : \text{SO}(3)^3 \rightarrow \mathbb{R}$, invariant under the diagonal action, with $\sum_{x \in X^3} K(x) \geq 0$ for every finite $X \subseteq \text{SO}(3)$.

Each $\mathcal{D}_\pi$ is invariant under the diagonal action, and $\mathcal{D}_\pi/\text{SO}(3)$ has dimension 0, 3 and 6 for $|\pi| = 1, 2, 3$.

Theorem 15.6 (Three-point bound). *Let K be a three-point positive kernel and let $(c_\pi)_\pi$ be real numbers, indexed by the set partitions of $\{1, 2, 3\}$, such that $K \leq c_\pi$ on $\mathcal{D}_\pi$ for every π . Put*

$$p_K(N) = \sum_{\pi} c_\pi (N)_{|\pi|}. \quad (48)$$

Then $p_K(N) \geq 0$ for every $N \leq N_{\max}(\tau)$. Consequently, if N_0 is an integer with $p_K(N) < 0$ for all integers $N > N_0$, then $N_{\max}(\tau) \leq N_0$.

Proof. Let X be non-overlapping with $|X| = N$. Subsets of non-overlapping sets are non-overlapping, so every $N \leq N_{\max}(\tau)$ occurs. If $x \in X^3$ and i, j lie in different blocks of $\pi(x)$, then $x_i \neq x_j$ are distinct elements of X , so $x_i^{-1}x_j \in \mathcal{D}$ by Definition 2.4. Hence $X^3 \cap \mathcal{D}_\pi = \{x \in X^3 : \pi(x) = \pi\}$, and this set has $(N)_{|\pi|}$ elements, one for each injective assignment of elements of X to the blocks of π . Therefore

$$0 \leq \sum_{x \in X^3} K(x) = \sum_{\pi} \sum_{x \in X^3 \cap \mathcal{D}_\pi} K(x) \leq \sum_{\pi} c_\pi (N)_{|\pi|} = p_K(N). \quad \square$$

If K is a three-point positive kernel, so is its symmetrization over the six permutations of the coordinates, with the same sum over X^3 . One may therefore assume K symmetric, in which case c_π depends only on the block sizes of π , leaving three constants $c_3, c_{2,1}$ and $c_{1,1,1}$, and

$$p_K(N) = c_3 N + 3 c_{2,1} N(N-1) + c_{1,1,1} N(N-1)(N-2). \quad (49)$$

Kernels come from the following construction, the group version of [1].

Proposition 15.7. *Let $\varphi : \text{SO}(3) \rightarrow \mathbb{R}^n$ be continuous and let $A \in \mathbb{R}^{n \times n}$ be symmetric positive semidefinite. Then*

$$K(h, s, s') = \varphi(h^{-1}s)^\top A \varphi(h^{-1}s'), \quad h, s, s' \in \text{SO}(3),$$

is a three-point positive kernel.

Proof. Invariance is built in. For finite X put $v(h) = \sum_{s \in X} \varphi(h^{-1}s)$. Then

$$\sum_{h \in X} \sum_{s, s' \in X} K(h, s, s') = \sum_{h \in X} v(h)^\top A v(h) \geq 0. \quad \square$$

Taking $\varphi(x)$ to be the vector of all entries of $T^1(x), \dots, T^L(x)$ and A block diagonal recovers the two-point kernels of Section 7 as functions of $s^{-1}s'$ alone. A general A is an arbitrary positive semidefinite matrix of size $n_L = \sum_{\ell \leq L} (2\ell+1)^2$, which is 35, 84, 165 for $L = 2, 3, 4$, against $\sum_{\ell \leq L} (2\ell+1) = 8, 15, 24$ for the two-point matrices C_ℓ . This is where the three-point bound gains, and where its cost lies. The certificate approach of Sections 9–13 carries over: the conditions $K \leq c_\pi$ on the strata become polynomial inequalities in two quaternionic variables instead of one, and Theorem 13.2 applies with the monomial basis in both variables.

15.3. A sharp three-point certificate. A certificate at a separation $\tau > \sqrt{2}$ proves a decimal of R_4 . The equality $R_4 = 1 + \sqrt{2}$ requires a certificate at $\tau = \sqrt{2}$ that is sharp in the following sense, the three-point analogue of Theorem 15.1. Write $\mathcal{D}_\pi(\sqrt{2})$ for the strata of Definition 15.5 at $\tau = \sqrt{2}$, and for a triple $(x_1, x_2, x_3) \in \mathcal{D}_{\{1\}\{2\}\{3\}}(\sqrt{2})$ call $\text{dist}(\ell_0, x_i^{-1}x_j \cdot \ell_0)$, $i < j$, its three distances. On this stratum each of them is at least $\sqrt{2}$.

Theorem 15.8. *Let K be a three-point positive kernel and let c_π , π ranging over the five set partitions of $\{1, 2, 3\}$, be real numbers such that*

- (i) $K \leq c_\pi$ on $\mathcal{D}_\pi(\sqrt{2})$ for every π ;
- (ii) $p_K(4) = \sum_\pi c_\pi(4)_{|\pi|} = 0$;
- (iii) *the level set $Z_3 = \{x \in \mathcal{D}_{\{1\}\{2\}\{3\}}(\sqrt{2}) : K(x) = c_{\{1\}\{2\}\{3\}}\}$ contains no triple whose three distances all exceed $\sqrt{2}$.*

Then $R_4 \leq 1 + \sqrt{2}$, and consequently $R_4 = 1 + \sqrt{2}$.

Proof. Suppose four pairwise non-overlapping cylinders of radius $\omega > 1 + \sqrt{2}$ touch the unit ball. By Section 2 there are distinct $g_1, \dots, g_4 \in \text{SO}(3)$ with $\text{dist}(\ell_0, g_i^{-1}g_j \cdot \ell_0) \geq \tau(\omega) = \frac{2\omega}{1+\omega} > \sqrt{2}$ for $i \neq j$. So $X = \{g_1, \dots, g_4\}$ is non-overlapping for $\tau = \sqrt{2}$ and all twelve of its distances are strictly larger than $\sqrt{2}$. As in the proof of Theorem 15.6, using (i) at $N = 4$ and then (ii),

$$0 \leq \sum_{x \in X^3} K(x) = \sum_\pi \sum_{x \in X^3 \cap \mathcal{D}_\pi(\sqrt{2})} K(x) \leq \sum_\pi c_\pi(4)_{|\pi|} = 0.$$

Hence $K(x) = c_{\pi(x)}$ for every $x \in X^3$. The 24 triples of distinct elements of X lie in Z_3 . Each of them has all three distances larger than $\sqrt{2}$, contradicting (iii). So $\mathcal{N}(\omega) \leq 3$ for every $\omega > 1 + \sqrt{2}$, and (2) gives the reverse inequality. $\square$

Hypothesis (ii) says that the three-point bound at $\tau = \sqrt{2}$ is exactly 3, with no slack. Hypothesis (iii) is the equality structure, and it is weaker than its two-point counterpart in two respects. First, it constrains the level set on the triple stratum only, and asks merely that every extremal triple contain one pair at distance exactly $\sqrt{2}$. The two-point hypothesis (iii) of Theorem 15.1 implies it, by Lemma 15.3, but a three-point kernel may satisfy it while being non-extremal on many pairs at distance $\sqrt{2}$ that occur in no extremal triple. Second, the level set of a sharp kernel is dictated by complementary slackness. At an optimum of the semidefinite program the dual object is a positive measure on the strata, and K is extremal exactly on its support. So (iii) holds as soon as the dual measure on the triple stratum is supported on triples having a pair at distance $\sqrt{2}$, for instance on the orbits of the two triple types of the ring of four, with distance patterns $(\sqrt{2}, \sqrt{2}, 2)$ and $(\sqrt{2}, 2, \sqrt{2})$ up to order. As in Remark 15.4, this much is also necessary: feeding the ring of four into the equality argument forces every one of its triples into Z_3 . So a sharp three-point kernel is extremal on the ring's triples, and (iii) asks only that it be extremal on nothing worse.

Theorem 15.8 is an outline of the three-point route. Its three hypotheses are what to test on a numerical solution of the three-point program at $\tau = \sqrt{2}$, and if they hold, the certificate framework of Sections 9–13 applies with two quaternionic variables. Whether a sharp three-point kernel exists we do not know. The size of the project, with matrices of order n_L rather than $\sum_{\ell \leq L} (2\ell + 1)$, puts it beyond what we can attempt at the moment.

16. OPEN PROBLEMS

- (1) *The case $n = 8$.* Decide Question 1.1 for $n = 8$.

The search of Section 14.1, which found a witness for every n in the window, found none for $n = 8$, and we expect that $(8, \omega_8) \in \mathcal{P}_{\text{cyl}}$. A proof would require a bound with target 8 at a separation just below $2 \sin \frac{\pi}{8}$, and the three-point bound of Section 15 is the first candidate.

- (2) *The second decimal of R_4 , and Conjecture 1.3.* Find a certificate with target 3 at a separation $\tau \in (\sqrt{2}, \frac{242}{171})$, or a sharp three-point certificate at $\tau = \sqrt{2}$ in the sense of Theorem 15.8.
- (3) *The asymptotic constant.* For $S \in (0, 1)$ and $T > 0$ let $\mu(S, T) = \min(2\Phi, (1 + a^2)/\sqrt{c_0})$ with Φ, a, c_0 as in Section 4. The proof of Corollary 1.8 gives $\liminf_{\omega \rightarrow 0^+} \omega \mathcal{N}(\omega) \geq \pi \mu(S, T)$ for every such pair, and numerically $\sup_{S, T} \mu(S, T) \approx 1.0546$. Determine this supremum exactly, and more ambitiously the true value of $\liminf_{\omega \rightarrow 0^+} \omega \mathcal{N}(\omega)$.
- (4) *An upper bound.* Is $\limsup_{\omega \rightarrow 0^+} \omega \mathcal{N}(\omega) < \infty$, i.e. $\mathcal{N}(\omega) = O(1/\omega)$? A positive answer with a good constant, combined with (3), would begin to pin down the extremal geometry of many thin cylinders touching a ball.

APPENDIX A. ELEMENTARY ESTIMATES

This appendix proves the estimates (E1)–(E5) of Section 4, used in Lemmas 4.1, 4.2, 5.1 and 6.4.

Lemma A.1. *Statements (E1)–(E5) hold.*

Proof. (E1): $f(x) = \sin x - x + \frac{x^3}{6}$ has $f(0) = 0$ and $f'(x) = \cos x - 1 + \frac{x^2}{2} \geq 0$, since $g(x) = \cos x - 1 + \frac{x^2}{2}$ has $g(0) = 0$ and $g'(x) = x - \sin x \geq 0$. The upper bound is standard.

(E2): $\frac{\tan y}{y}$ is increasing on $(0, \frac{\pi}{2})$ because $(y \sec^2 y - \tan y) = \frac{y - \sin y \cos y}{\cos^2 y} > 0$. Hence for $y \in (0, 1]$, $\tan y \leq y \tan 1 < y\sqrt{3}$, using $1 < \frac{\pi}{3}$ and the monotonicity of $\tan$. Then $h(y) = y + y^3 - \tan y$ has $h(0) = 0$ and $h'(y) = 3y^2 - \tan^2 y > 3y^2 - 3y^2 = 0$.

(E3): apply $\tan$ to both sides for the identity. The condition $pq > -1$ keeps the difference in $(-\frac{\pi}{2}, \frac{\pi}{2})$. The inequality $\arctan z \leq z$ holds since $\tan$ is convex on $[0, \frac{\pi}{2})$.

(E4): the derivative has numerator $1 + a^2 + z^2 > 0$.

(E5): squaring, the claim is $(1 + z^2)^2(1 - z^2) \geq 1$, i.e. $z^2(1 - z^2 - z^4) \geq 0$, which holds when $z^4 + z^2 \leq 1$, in particular for $z \leq \frac{3}{4}$, as $\frac{81}{256} + \frac{9}{16} < 1$. $\square$

APPENDIX B. DEPOSITED DATA AND STANDALONE VERIFIER

The data are in plain-text files in a self-describing format, designed to be read by a short standalone parser. Each record consists of a key, written `[key]Name[/key]`, followed by a value block `[value] ... [/value]`. A value block holds one or more rows. Each row, and each entry within a row, is delimited by `{` and `}`. Every value is therefore a nested list of integers of depth two, including those that carry a single number, which appear as one row with one entry. Keys that depend on an index ℓ or a end in it as a four-digit number.

The *universal* records, identical in every certificate file, are, for each $\ell = 1, \dots, L$:

- `WNum0000 $\ell$` and `WDen0000 $\ell$` : the weights w_i^ℓ as numerator/denominator pairs, forming

$$D_\ell = \text{diag}(w_1^\ell, \dots, w_{2\ell+1}^\ell);$$

- `Tpoly0000 $\ell$` together with the scale `Tden0000 $\ell$` . Each row has 7 numbers $i, j, e_0, e_1, e_2, e_3, c$. These seven numbers mean that the polynomial $T_h^\ell(q)_{ij}$ contains the term $(c/T\text{den}_\ell) q_0^{e_0} q_1^{e_1} q_2^{e_2} q_3^{e_3}$. The polynomial $T_h^\ell(q)_{ij}$ is the sum over the rows carrying that i, j .

Each *certificate* file records:

- `Lmax`: the truncation L . All degree-dependent constants of the verifier are derived from it;
- `targetBound`: the positive integer N of the claim. It enters the verification only through the counting check (V6);
- `tauSqNum` and `tauSqDen`: the positive integers A and B with $\tau^2 = A/B$;
- `UhPoly` and `QhPoly`: the integer coefficient lists of the generators U_h and Q_h , in the row format of `Tpoly`. The verifier does not construct these polynomials; it validates the deposited lists against the closed forms (38)–(39) at the deposited A and B (check (V2) below);
- `MNum`, `MDen`, `TstarNum`, `TstarDen`: the constants M and T^* ;
- `cDenLog2`: the even integer $\kappa \in [0, 60]$ scaling the next record;
- `Cscaled0000 $\ell$` = $2^\kappa C_\ell$, as the rows of an integer matrix;
- `Swit0000 $\ell$` and `Wwit0000 $\ell$` : the integer scalar S_ℓ and the integer matrix W_ℓ of Proposition 12.2;
- `sosExponentK`: the integer k of the identity (45);
- `sosDenLog2`: the even integer κ_s scaling the next record;
- `sosGram0000 $a$` = $2^{\kappa_s} \Sigma_a$ for $a = 0, \dots, 3$, as the rows of integer symmetric matrices of the sizes prescribed by Theorem 13.2;
- `sosSwit0000 $a$` and `sosWwit0000 $a$` : the integer scalar S_a and the integer matrix W_a of Proposition 12.2 for Σ_a .

Every other quantity in the proof is derived from these records: the reciprocals $1/w_i^\ell$ from `WNum/WDen`, and the polynomials P and $G = M(|q|^2)^L - P$ from the C_ℓ , the tables and the weights. The verifier constructs no polynomial in $q_0, \dots, q_3$ other than by expanding deposited records.

On these records the standalone verifier confirms, for each certificate in turn:

- (V1) the claim is well formed: `targetBound` ≥ 1 , $M < 0$, and $0 < \tau^2 < 4$, so that $\omega = \tau/(2 - \tau)$ is defined and positive;
- (V2) the deposited generator records `UhPoly` and `QhPoly` coincide, coefficient by coefficient, with the closed forms (38)–(39) evaluated at the deposited A and B ;

- (V3) (H1) and the positivity of the weights, by inspection of the exponents in $\text{Tpoly000}\ell$ and of the signs in $\text{WNum000}\ell$, $\text{WDen000}\ell$; (H2) and (H3), by expanding both sides of the two identities of Section 12.1 and comparing them coefficient by coefficient (once, since the tables are universal);
- (V4) the hypothesis of Proposition 12.2, forming $E_\ell = S_\ell(2^\kappa C_\ell) - W_\ell^\top W_\ell$ and testing symmetry and diagonal dominance;
- (V5) (H5), by evaluating $\sum_\ell \sum_i (C_\ell)_{ii}/w_i^\ell$ and comparing it with the deposited T^* ;
- (V6) the *counting inequality*

$$T^* < (-M) \cdot \text{targetBound}, \quad \text{equivalently} \quad \text{targetBound} \geq 1 + \left\lfloor \frac{T^*}{-M} \right\rfloor,$$

in exact rational arithmetic, which by Theorem 10.2 turns hypotheses (H1)–(H6) into the bound $N_{\max}(\tau) \leq \text{targetBound}$;

- (V7) the hypothesis of Proposition 12.2 for each multiplier matrix Σ_a , from the records $\text{sosWit000}a$ and $\text{sosWit000}a$, establishing $\Sigma_a \succeq 0$ (step (S2) of Section 13);
- (V8) (H6), by expanding both sides of the identity (45) and comparing them coefficient by coefficient (steps (S1), (S3) and (S4) of Section 13); the conclusion follows by Theorem 13.2.

Every one of these checks is exact. The verifier uses no floating-point arithmetic. Integers are held in fixed-width machine words of 128, 256 or 512 bits, selected from the deposited L . Each operation is checked for overflow. For rational numbers, we use two representations. If the number is not dyadic, then we store the numerator and denominator. In the dyadic case the denominator is a power of two, and we keep only its exponent. Addition, multiplication and comparison are implemented so that no rounding can occur, and every operation is guarded by an overflow test. Any guard that fires sets a permanent flag; a flagged run reports the fact and its output is discarded.

The verifier’s contract. The verifier is a single program with a one-line verdict. Invoked without arguments it prints its own contract, which we reproduce in substance:

The input file contains, among other data, a positive integer `targetBound` and positive integers `tauSqNum`, `tauSqDen` with `tauSqNum < 4 · tauSqDen`. Define $\tau = \sqrt{\text{tauSqNum}/\text{tauSqDen}}$ and $\omega = \tau/(2 - \tau)$. If the last line printed by the program is `THE THEOREM WAS PROVED`, then the following statement is a theorem: *no more than targetBound pairwise non-overlapping cylinders of radius ω can simultaneously touch the unit ball*. Everything else in the input file is a machine-generated witness. No modification of the input, however adversarial, can make the program print the verdict for a false statement; the only realistic effect of tampering is that the proof is not found.

In the language of Definition 1.4: the verdict certifies $(\text{targetBound}, \omega) \in \mathcal{P}_{\text{cyl}}$.

The final line is printed only if all of (V1)–(V8) succeed and every arithmetic flag is still clear at the end of the run. Any failure, at any stage, suppresses it. The soundness of the verdict therefore rests on the verifier source, the compiler and hardware executing it, and Theorem 10.2. Two input switches, `skipMultVerification` and `skipOrthVerification`, allow the universal checks in (V3) to be skipped when the tables have been verified separately. If either switch is active, the verdict is conditional on those identities. Both switches are off in all five deposited certificates.

The window configurations of Section 6 are deposited in the format described there, and are checked together with the inequalities of Table 1 by the run `cylie -i` constructions of Section 4.4.

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ON THE USE OF AI

Ideas were discussed and the literature was reviewed with the help of Anthropic’s Claude models. The programs that searched for certificates and for the configurations of Section 6 were written with the assistance of these models. The verifier [11] was written by the authors and reviewed and tested both by the authors and by the models. The proofs use the certificates and configurations only through the verifier, and do not depend on how they were found. The text was written by the authors and then proofread and edited with the same models.

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